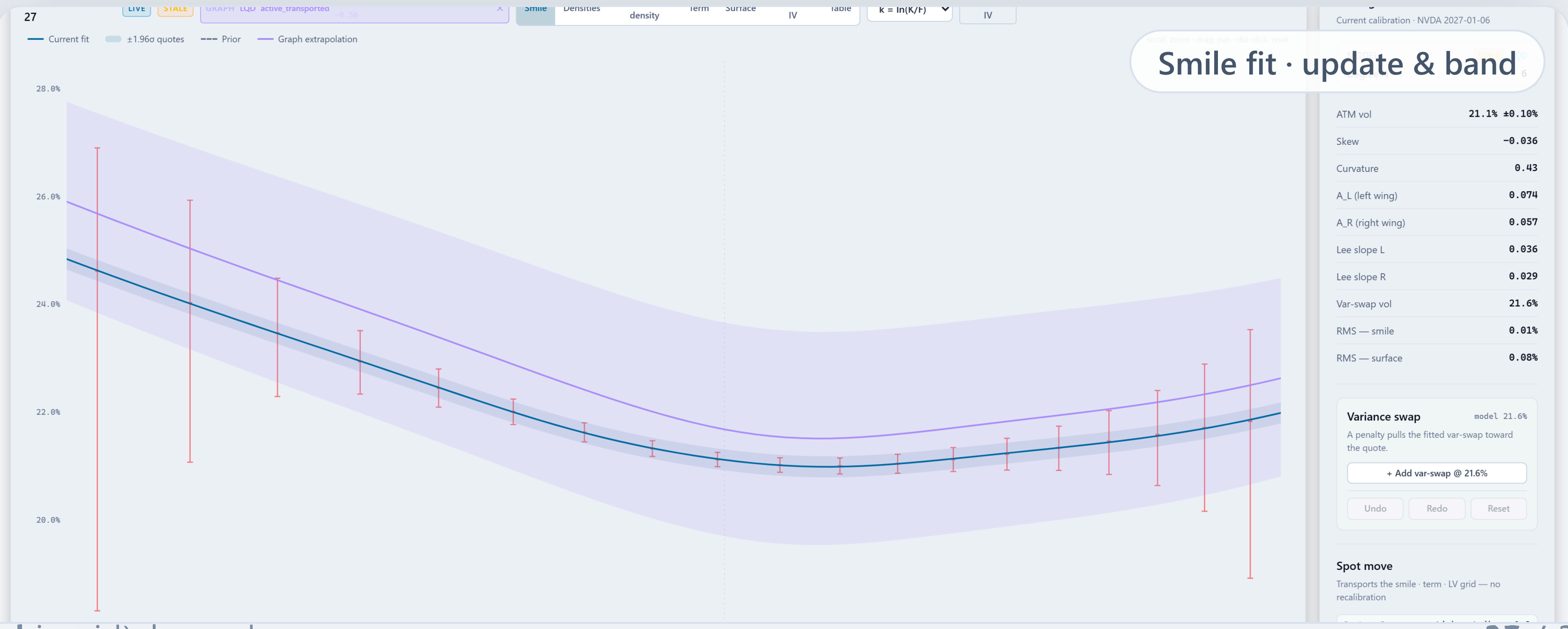
VOL-FITTER

Certified volatility surfaces, with graph propagation to the smiles nobody quotes.

A volatility workstation that turns sparse, noisy option markets into certified, arbitrage-clean surfaces with honest uncertainty — and propagates liquid-name information across expiries and assets, with exact attribution.

- **4** smile models, one calibration spine
- **1,635** tests green (1,506 backend · 129 UI)
- **22**-case certification pack, client-facing
- backtested on **3** regimes + an intraday campaign



THE PROBLEM WHY OPTION SURFACES ARE HARD

Listed option markets are sparse, noisy, and structurally constrained — most of the surface is never quoted.

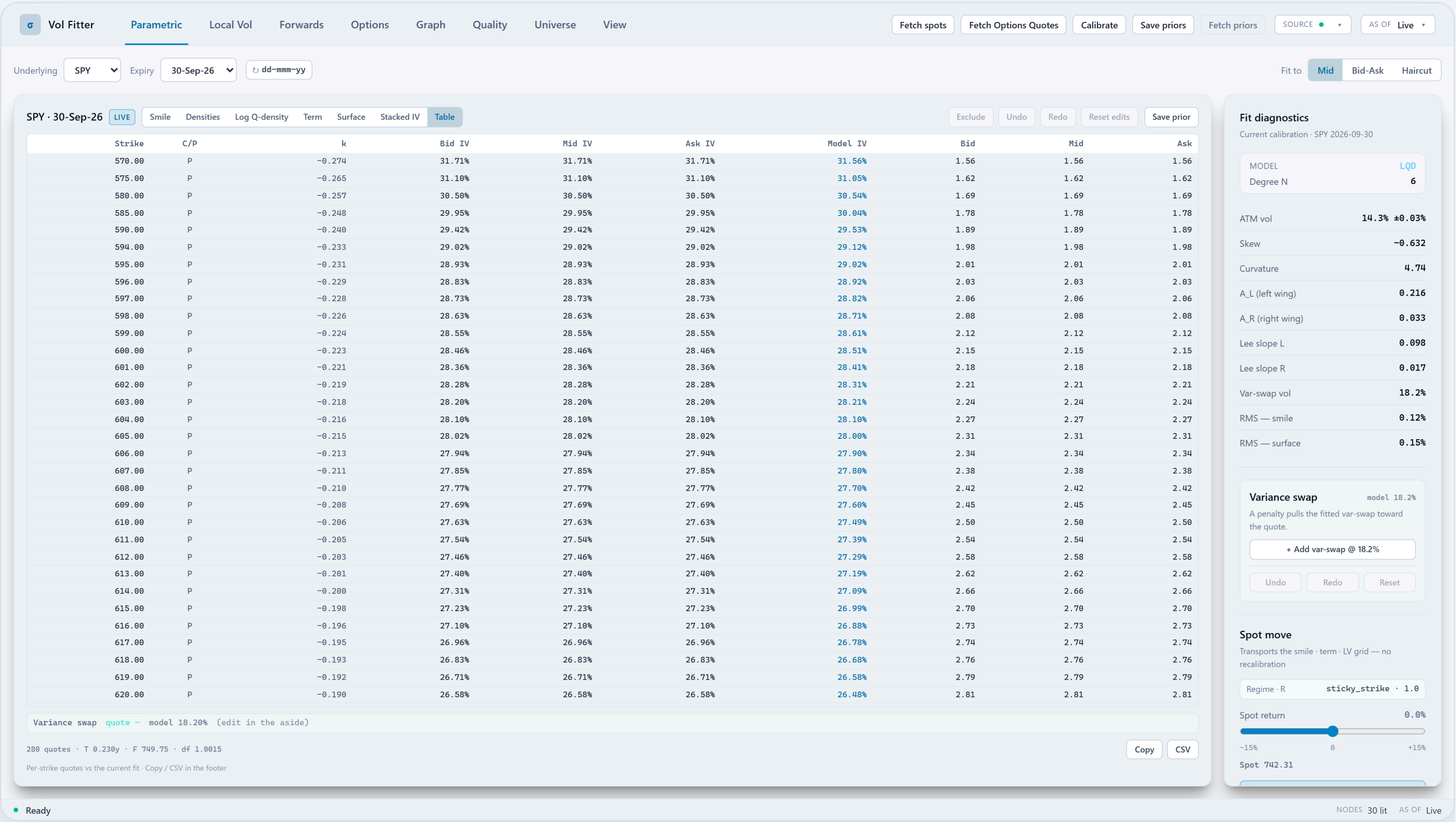
- **Wide, asymmetric spreads** — wings quote at multiples of ATM spread; mids are not fair value.
- **American exercise** — listed equity quotes embed an early-exercise premium that poisons European-model fits.
- **Structural no-arbitrage** — butterfly and calendar constraints couple every strike and expiry; a fit that ignores them produces tradable-looking nonsense.
- **Events distort the clock** — earnings compress into a day what diffusion spreads over weeks.
- **Most names are dark** — beyond the top tickers, whole expiries have no usable quotes at all.

most

of the (underlying × expiry) universe has no reliable quotes at any moment

every

downstream decision — pricing, hedging, alpha — still needs the full surface



The raw material — live SPY chain (Massive feed). Per strike: bid / mid / ask implied volatility (IV), the model's fitted IV, and the deviation. The fitter's job is to decide which of these numbers to believe, and by how much.

TRADING RELEVANCE

Trading desks need a system that turns **partial, contradictory evidence into a full, arbitrage-clean surface with uncertainty**.

ARCHITECTURE ONE PIPELINE, FOUR SMILE MODELS

One quote pipeline feeds four smile models — all speaking the same handle vocabulary.

Every quote passes through one deterministic pipeline, and every model reads and writes the same small set of smile descriptors — the **"handles"**: at-the-money (ATM) volatility, skew, curvature, wing slopes. That shared vocabulary is what lets the later layers (priors, filtering, the graph) compose.

Fetch — Yahoo · Bloomberg · Massive · Synthetic, live or as-of → **De-Americanize** — CRR inversion, Numba kernel → **Forwards** — parity regression, clamped & de-biased → **Prepare** — weights, bands, event clock → **Calibrate** — LQD · SVI-JW · MCS · Local Vol → **Persist & filter** — Bayesian priors, Kalman filter → **Graph** — propagate to dark nodes

THE SHARED PRICING PRIMITIVE — NORMALIZED BLACK ON TOTAL VARIANCE

$$w(k,T) = \sigma^2_{\mathrm{BS}}(k,T)\,T, \qquad \mathrm{Bl}(k,w) = \Phi(d_+) - e^k\,\Phi(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}$$

B(k,w) — normalized (forward-units, undiscounted) Black call; k — log-moneyness $\ln(K/F_T)$; w — total implied variance $\sigma^2_{BS}\cdot T$; $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$; Φ — standard normal CDF.

- All models quote the same objects: **total variance w(k), ATM vol, skew, curvature, wing scales**.
- Because priors, the filter and the graph act on the **same handles**, they compose instead of fighting.
- **Every feature is strictly additive** — bands, var-swaps, events, priors, calendar coupling: byte-identical to OFF when disabled, and that byte-identity is test-locked.

The four smile models

SVI-JW	the industry benchmark — Gatheral's SVI ("Stochastic Volatility Inspired") in jump-wing trader units; the reference every accuracy claim here is measured against
LQD	the house model — **Log-Quantile-Density**: density-based (it parametrizes the return distribution itself), fits complex shapes, natively arbitrage-free in strike space, exact ATM handles
MCS	**Multi-Core Sigmoid** — a classical kernel-based implied-variance model: a sigmoid base plus localized kernels, reaching complex shapes (event "W"s) a convex model cannot
Local Vol	fully non-parametric — a piecewise-affine local-variance grid calibrated to observed option prices through the pricing PDE, driving the surface directly without modeling the IV curve

- **Every guarantee is graded and executable** — structural where the model provides it (LQD, Local Vol), **certified at publish** where it does not (SVI/MCS carry a dense-grid butterfly certificate; an uncertified slice cannot become a mark), measured everywhere.

TRADING RELEVANCE

The architecture is why the headline feature works at all: **graph propagation is only possible because everything upstream normalizes into one coordinate system**.

THE CORE IDEA, WORKED   ONE END-TO-END EXAMPLE

The end goal first: SPY reprices — an unquoted NVDA smile follows, with uncertainty attached.

A controlled demo session (synthetic feed, so it is exactly reproducible): only SPY is quoted ("lit"); QQQ, AAPL, NVDA and IWM are dark. SPY's whole surface reprices +150 bp. **The rest of this deck exists to earn each arrow in this picture.**

1 · LIT MOVE

Every SPY quote mid is amended **+150 bp**, then SPY's four expiries are genuinely recalibrated on the shifted quotes — **real fits, not typed shifts**

2 · INNOVATION

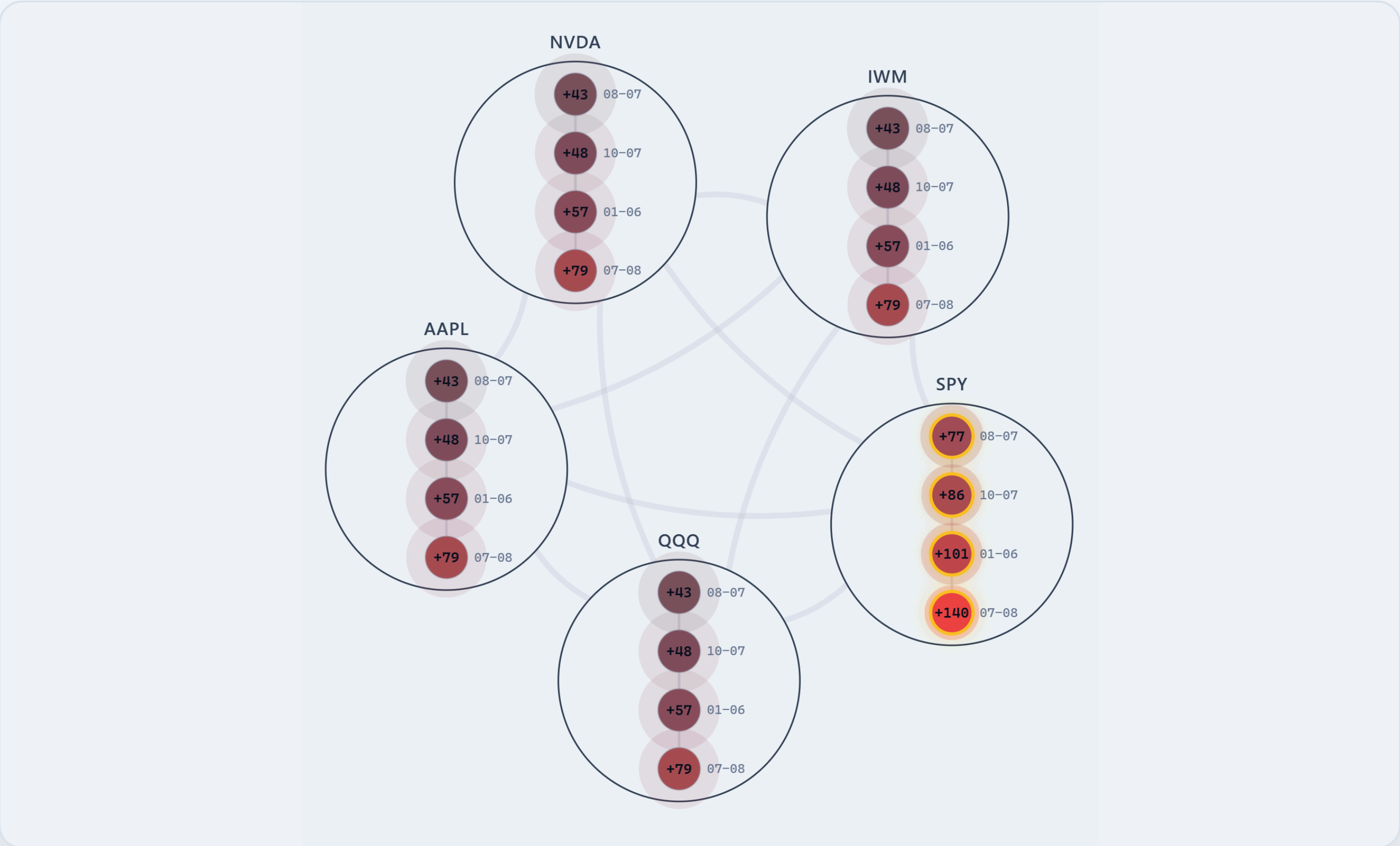
Innovation = new fit − transported prior, per handle: **+97…+151 bp of ATM news** across SPY's expiry ladder. Only this **change** propagates — never vol levels

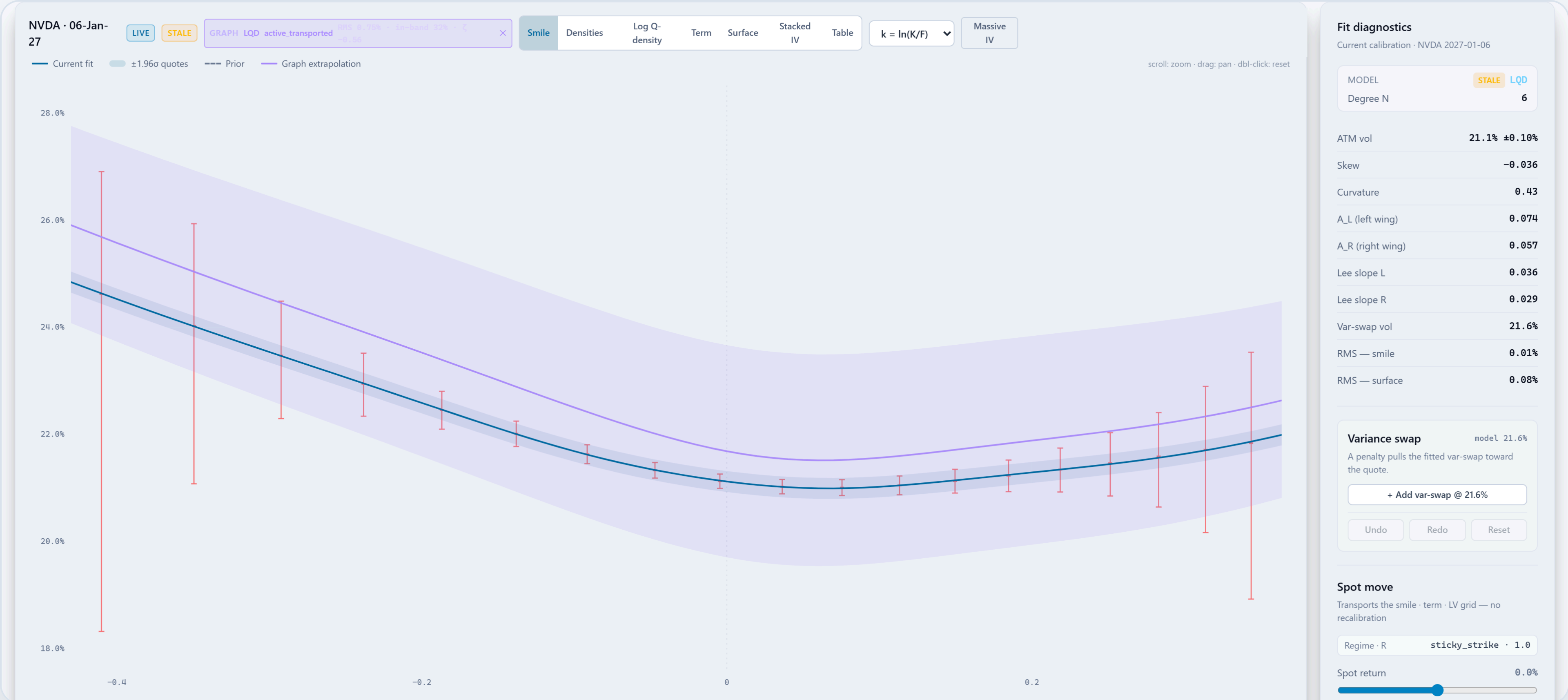
3 · PROPAGATION

The news diffuses along calendar and cross-asset edges; each dark node blends it with its own prior by **Bayesian precision weights** — a deliberate fraction, with widened uncertainty

4 · DARK POSTERIOR

NVDA's unquoted smiles lift **+43…+79 bp of ATM vol** (≈½ the SPY news), inside a **±1.9–2.0 vol-pt** 95% credible band (±1.96σ) they can be judged against later





WHY IT MATTERS

The point of the exercise: **the unquoted majority of the surface universe stays marked, moves for stated reasons, and carries stated uncertainty**. The rest of the deck documents how each step is built and where it is validated.

VOCABULARY EIGHT WORDS THE DECK USES PRECISELY

Eight terms, defined once — everything after this slide uses them exactly.

Handle

A smile descriptor a desk quotes in: **ATM vol, skew, curvature, wing slopes**. In this app they are model parameters (or exact functions of them), not chart read-offs.

Node · lit / dark

One (underlying, expiry) smile — the unit of everything here. **Lit** = has usable quotes, gets calibrated; **dark** = unquoted, gets reconstructed.

Transported prior

Yesterday's fitted surface, moved to today's spot by the chosen spot-vol rule (SSR) — the **"no news" baseline** for every node.

Innovation

Today's calibration **minus** the transported prior, per handle — the genuine news in a lit node. The only thing the graph propagates.

Posterior · credible band

The estimate after combining sources (prior + news), with its uncertainty. A dark node's posterior comes with a band it can be **judged against later**.

RMS · bp

Fit error = **root-mean-square implied-vol error**, in vol basis points (1 bp = 0.01 vol point). 10 bp ≈ tighter than most quoted spreads.

ζ (standardized residual)

Error divided by its own predicted uncertainty — computed wherever the system quotes a band: graph reconstructions, the LOO backtest, the observation filter. If the bands are honest, ζ ≈ standard normal: **mean 0 (unbiased), std 1 (calibrated)**.

Static no-arbitrage · certified

What "arbitrage-free" means here: **butterfly-clean within each slice, calendar-clean across expiries** — and **certified**: a dense-grid check on every fit, hard-gating publish. Not a claim about dynamic or model risk.

LQD, the house model: the density is positive by construction, and the ATM handles are exact.

LQD = Log-Quantile-Density. Instead of parametrizing the implied-vol curve and hoping the implied density stays positive, the model parametrizes the log of the quantile density of the terminal return — an object whose exponential is positive by definition. Positive density ⇔ no butterfly arbitrage, so admissibility is built in, not enforced.

LOG-QUANTILE-DENSITY BASIS — POSITIVE DENSITY BY CONSTRUCTION

$$\ell(u) = -\log u - \log(1-u) + (1-u)\,L + u\,R + \sum_{n=2}^{N} a_n\,P_n(1-2u)$$

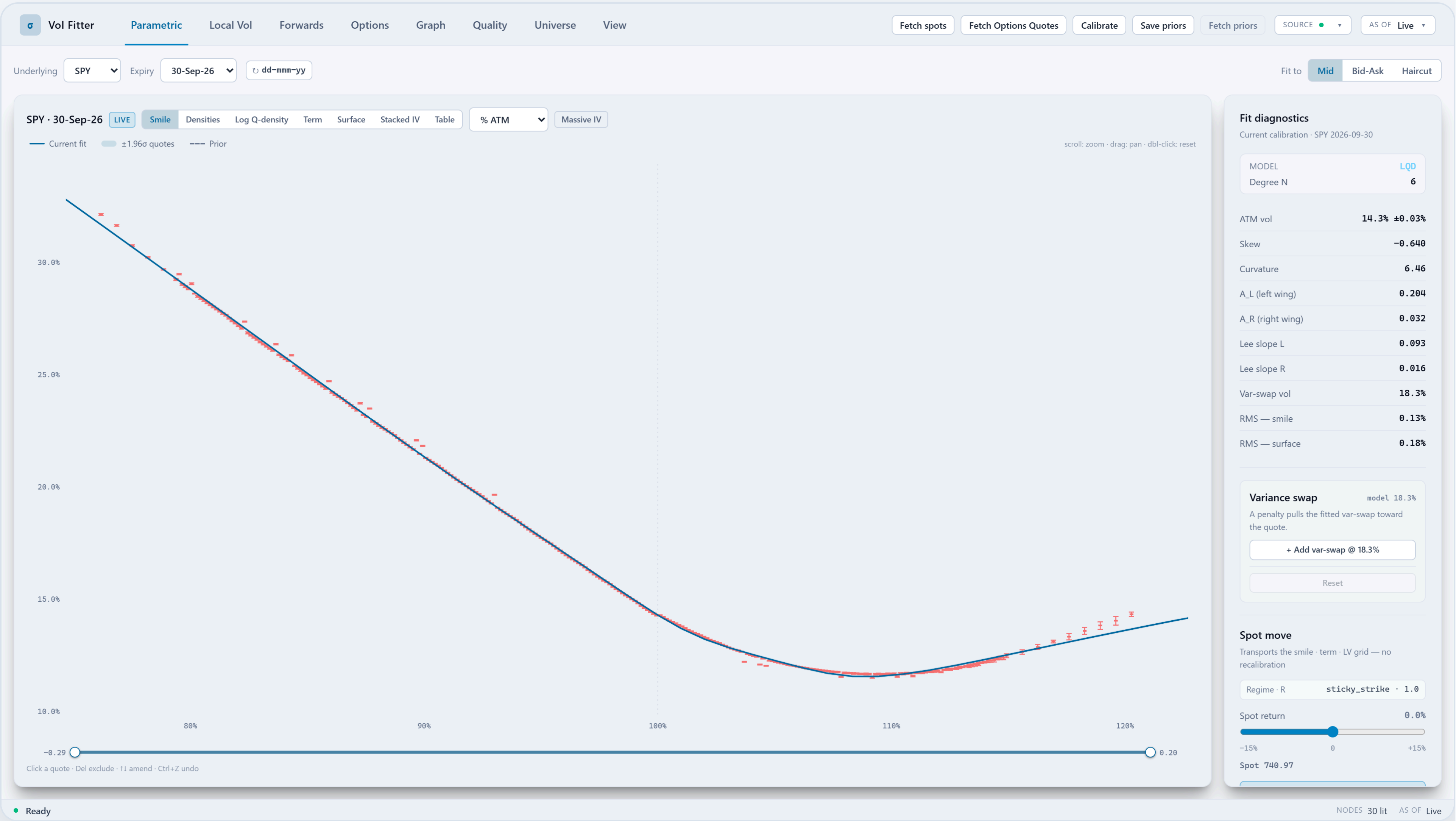
$\ell = \log q$ — log quantile density of the log-return ($q = Q'$, Q the quantile function); u — probability level in (0,1); L, R — left/right endpoint (tail-scale) coefficients; a_n — body coefficients; P_n — **Legendre polynomial of degree n** in $1-2u$; N — body order.

EXACT ATM HANDLES — VOL, SKEW, CURVATURE IN CLOSED FORM

$$\sigma_0 = \sqrt{\frac{w_0}{T}}, \qquad s_0 = \frac{w_1}{2\sqrt{T\,w_0}}, \qquad \kappa_0 = \frac{w_2}{2\sqrt{T\,w_0}} - \frac{w_1^2}{4\sqrt{T}\,w_0^{3/2}}$$

σ_0 — ATM implied vol; s_0 — ATM skew $d\sigma/dk$; κ_0 — ATM curvature $d^2\sigma/dk^2$; w_0 — ATM total variance (solves $\mathrm{Black}(0,w_0) = C(0)$); w_1, w_2 — 1st/2nd log-strike derivatives of $w(k)$ at $k = 0$, exact by implicit differentiation; T — expiry.

- **Butterfly-free by construction** — the density is an exponential; no penalty needed within a slice.
- One hard constraint only: $A_R < 1$ (a finite forward) — the default chart maps the wall to infinity, $A_R = \mathrm{expit}(\rho)$ (adopted from a bank-committee review). Lee slopes come out structural.
- **Analytic Jacobian via implicit-root cancellation** — measured **1.4–2.0× vs finite differences** across degrees 6–12; a 13-parameter LQD fits in tens of milliseconds.



Parametric workspace — live SPY (Massive feed), strike axis in % of the ATM forward. Blue — the LQD fit; red marks — quoted bid/ask per strike. The aside reports the handles the model controls directly: ATM vol, skew, curvature, wing scales A_L/A_R, Lee slopes, var-swap level.

TRADING RELEVANCE

Handles are the desk vocabulary — **ATM, skew, curvature are model parameters, not read-offs**, so priors, filters and the graph all speak the same coordinates.

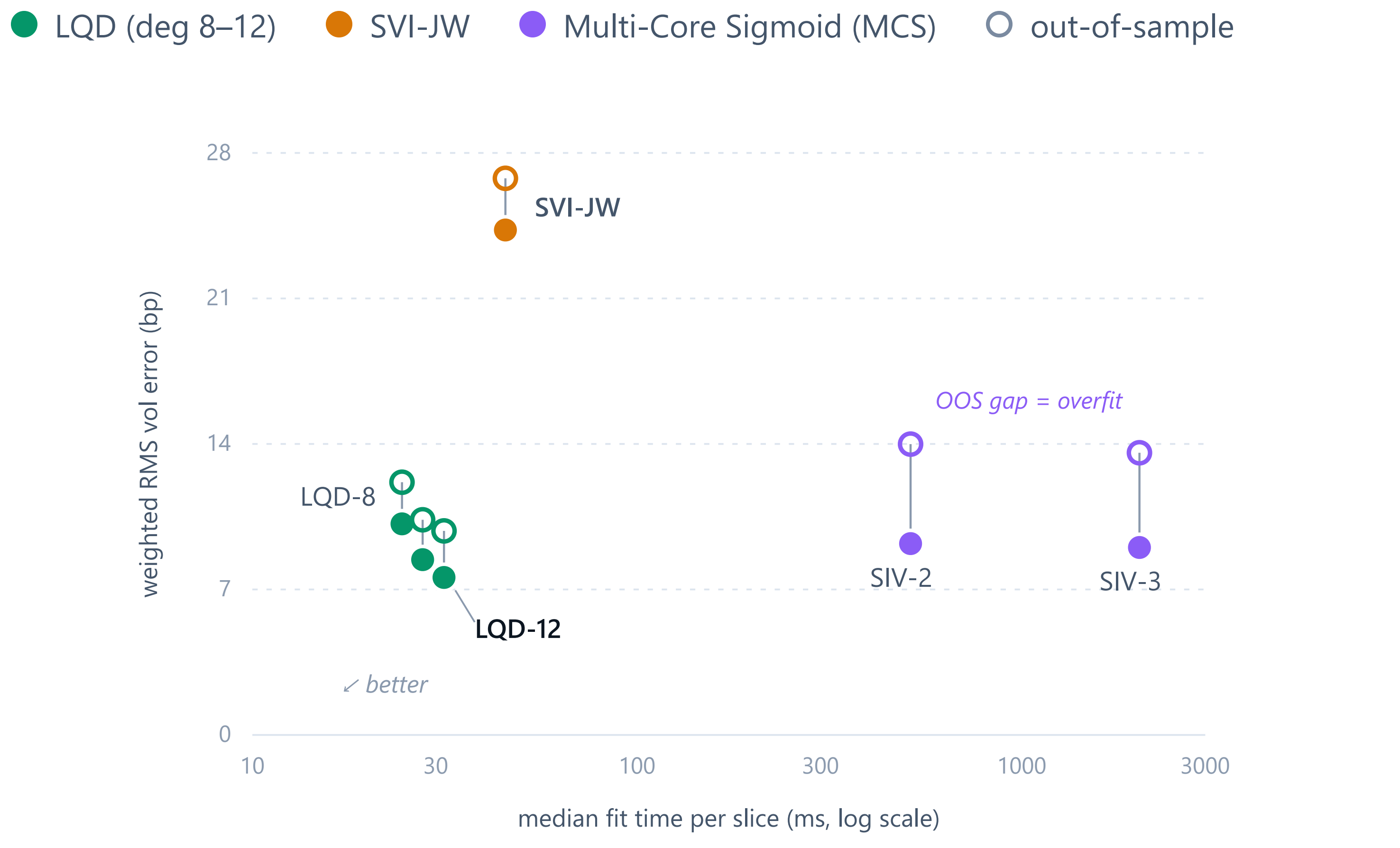
BACKTEST · 3 REGIMES VALIDATION

Backtested on real NBBO history: LQD-8…12 fit 2–3× tighter than SVI-JW; the default LQD-6 is ~1.6× tighter at twice SVI's speed.

Offline backtest: historical NBBO quotes (national best bid/offer) replayed through the production calibrators over three regimes, on the 8-asset pilot universe (SPX · NDX · RUT · EEM · EFA · AAPL · NVDA · JPM, full expiry ladders). **In-sample** = RMS vol error on the strikes used in the fit; **out-of-sample (OOS)** = error on held-out strikes — the overfitting test. The hold-out is every 3rd strike across the chain, the identical mask for every model — it samples ATM and wings alike and cannot favour one model's basis.

Fit error vs fit time — spike regime (Aug 2024), 1,576 nodes

Lower-left is better · filled = in-sample RMS · hollow = out-of-sample · times pre-date SVI's analytic Jacobian (now ~10 ms)



MODEL	IN-RMS BP	OOS BP	FIT MS	VS SVI
SVI-JW	24.3	26.8	45.5	1.00×
LQD-8	10.2	12.2	24.5	0.42×
LQD-12	7.6	9.8	31.5	0.31×
MCS-3	9.0	13.6	2023	0.37×

0.31–0.45×
LQD-12 in-RMS vs SVI, consistent across spike / high / low regimes

0
fit failures across all models and regimes

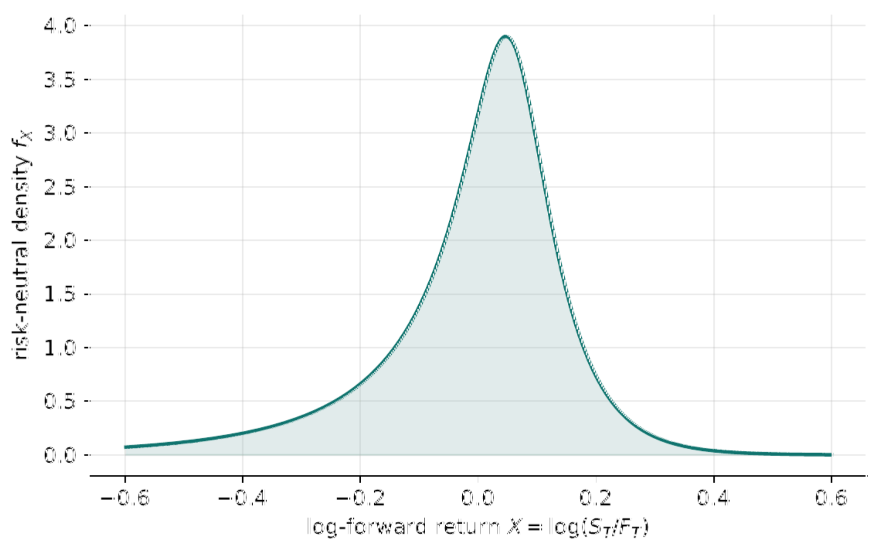
- Out-of-sample tracks in-sample for LQD — **no measured overfit over the tested degrees (6–12)**.
- MCS's extra cores buy in-sample error and **give it back out-of-sample** — production caps cores at 2 and regularizes the put wing.
- Fit times are as measured **in this sweep** — it ran before SVI's analytic Jacobian shipped (SVI is ~10 ms since). The **accuracy columns are the durable result**; on speed the honest claim is low tens of milliseconds, inside the 50 ms rail.

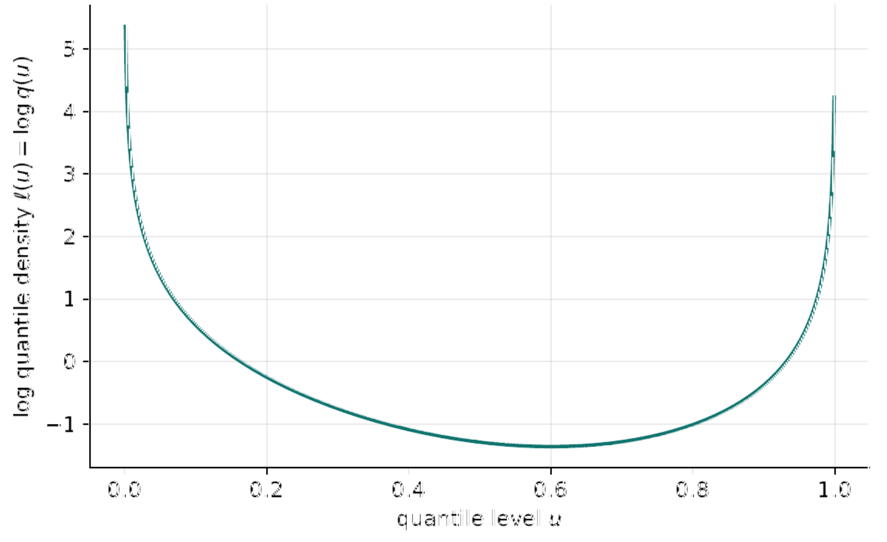
TRADING RELEVANCE

On replayed market history the house model fits **up to 2–3× tighter than the SVI baseline (degree 8–12; ~1.6× at the default degree 6) in low tens of milliseconds, with no measured overfit over the tested range** — and the same harness reports the arbitrage metrics alongside the errors.

An LQD fit is a probability law — its density and quantile views read straight off the model.

Any fitted smile implies a risk-neutral distribution of the terminal return. LQD (**Log-Quantile-Density**, the house model of the previous slides) parametrizes that distribution directly — its parameter *is* the log of the quantile density $\ell(u)$ — so the density and quantile views below are the model read in its native coordinates, not a numerical second derivative of a fitted IV curve.





Generated by the production LQD fitter. Left — the risk-neutral density behind a fitted LQD smile, non-negative everywhere by construction. Right — the log-quantile-density $\ell(u)$ itself, plotted against the quantile level u: the model's native coordinate, where each wing is one endpoint scale (A_L, A_R) and the body is a smooth Legendre expansion.

PRICING IDENTITY IN LOGIT COORDINATES — NO BLACK INVERSION IN THE LOOP

$$C(k) \;=\; A(z_k) \;-\; e^k\,(1-u_k)$$

$C(k)$ — normalized call at log-moneyness k; z_k — logit point solving $Q(z_k) = k$ (Q = log-return quantile); $u_k = \Lambda(z_k)$ — probability level (Λ = logistic function); $A(z)$ — upper asset-share integral $\int_z^\infty e^{Q(s)}\Lambda(s)(1-\Lambda(s))ds$.

STRUCTURAL TAILS — LEE SLOPES FROM THE ENDPOINT SCALES

$$A_R < 1, \qquad \beta_L = \psi(1/A_L), \quad \beta_R = \psi(1/A_R - 1), \qquad \psi(p) = 2 - 4\left(\sqrt{p^2+p} - p\right)$$

β_L, β_R — left/right asymptotic slopes of total variance $w(k)/|k|$; ψ — Lee's moment function $\psi(p) = 2-4(\sqrt{p^2+p}-p)$; p — critical moment exponent of S_T/F_T; A_L, A_R — LQD endpoint scales (quantile tail slopes; $A_R < 1 \Leftrightarrow$ finite forward).

- LQD prices come from a **single native asset-share quadrature** — quotes, densities, quantiles and var-swaps all read off the same fitted object.
- Tail **shape is a stated modelling choice** (exponential return tails); the tail **scales are fitted and diagnosed** — A_L, A_R and the implied Lee slopes are reported with every fit.

TRADING RELEVANCE

Because the density is the model, **digital prices, tail odds and var-swap strikes are consistent with the smile by construction** — not by post-processing.

SVI, the market's standard smile — spoken here in jump-wing handles, with the conversion domain made explicit.

SVI ("Stochastic Volatility Inspired", Gatheral) is the market-standard 5-parameter smile; most desks and vendors quote surfaces in it. The **jump-wings (JW)** reparametrization re-expresses the same curve in trader units: level, skew, two wing slopes, minimum.

RAW SVI — WHAT THE OPTIMIZER SEES

$$w(k) \;=\; a + b\left\{\rho\,(k-m) + \sqrt{(k-m)^2+\sigma^2}\right\}$$

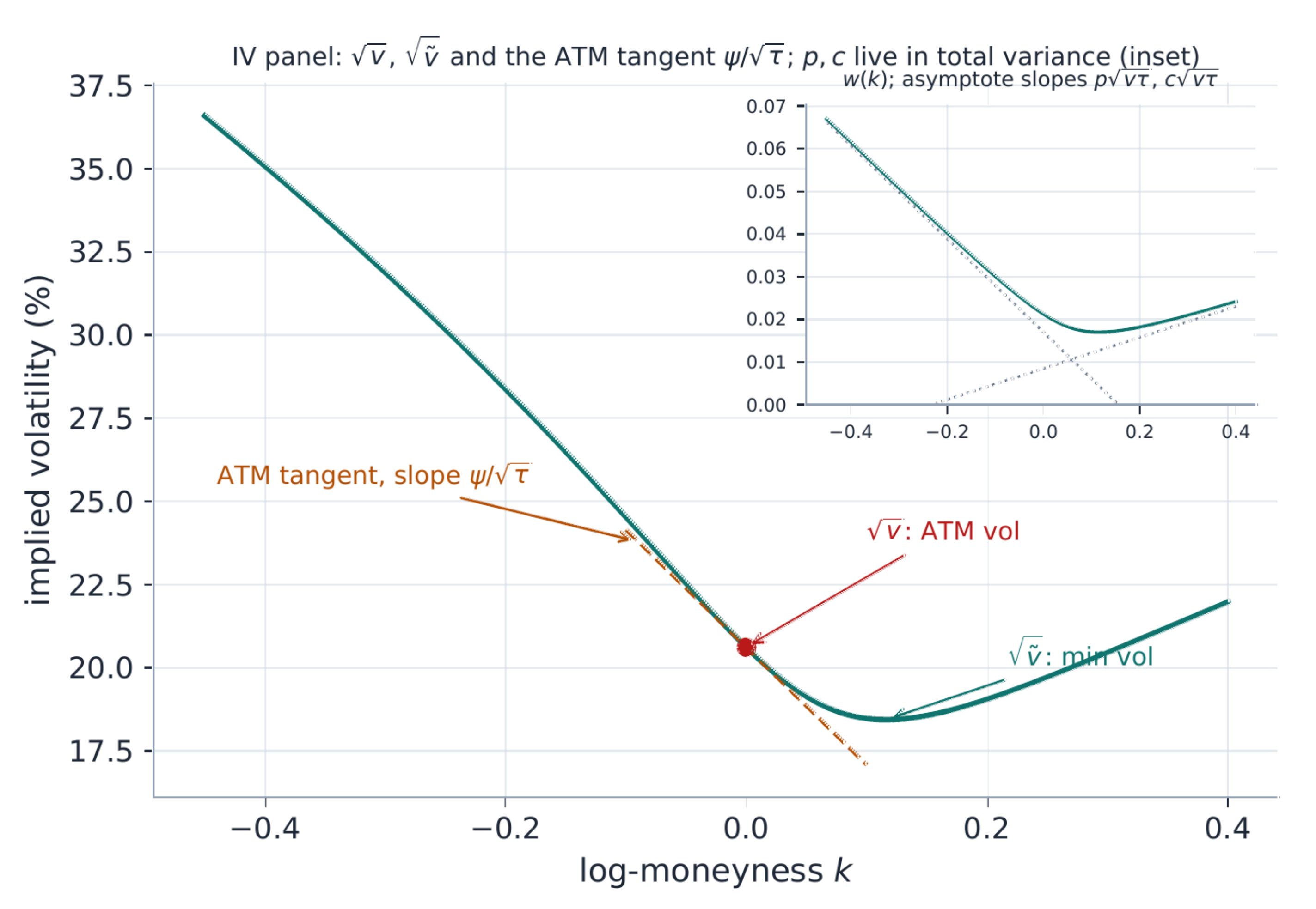
w(k) — total implied variance; k — log-moneyness ln(K/F); a — level; b — wing steepness; ρ — tilt; m — vertex shift; σ — vertex rounding.

JUMP-WING HANDLES — WHAT HUMANS SEE

$$v = \frac{w(0)}{t}, \quad \psi = \frac{b}{2\sqrt{w_0}}\left(\rho - \frac{m}{\sqrt{m^2+\sigma^2}}\right), \quad p = \frac{b\,(1-\rho)}{\sqrt{w_0}}, \quad c = \frac{b\,(1+\rho)}{\sqrt{w_0}}$$

v — ATM variance w(0)/t; ψ — ATM skew; p, c — put/call wing slopes; ṽ — minimum variance; t — expiry; w_0 = vt; a, b, ρ, m, σ — the raw parameters above.

- **Fitted in a structural chart** — lifted coordinates keep every iterate a valid hyperbola with wings *strictly inside* the Lee fence. Default by a two-round **pre-registered ~28,800-fit benchmark**: all 12 precision medians better-or-equal, zero breaks, ~3× faster (raw exhausted its eval budget on ~33% of fits).
- **Analytic Jacobians on both charts** — raw **26.3 → 10.2 ms (~2.6×)**; structural **2.1–2.4× again** at identical smiles — deriving it flushed out two latent float-boundary bugs.
- The raw↔JW conversion is exact **on a guarded domain**, with a **desk ticket layer**: 25Δ/10Δ RR/BF, forward bumps.



The five jump-wing handles, in honest units. The IV panel shows √v at the money, the minimum √ṽ, and the ATM tangent of slope ψ/√τ; the wing handles p, c are normalized *total-variance* slopes, on the w(k) inset. JW is the trader coordinate system — production fits the structural chart (raw = the explicit rollback).

TRADING RELEVANCE

Native fluency in the market's standard model — **fitted in a benchmark-ratified structural chart, with the domain-guarded JW conversion and a desk-unit ticket layer (RR/BF, forward bumps) shipped and test-locked**.

SVI cleanliness is not assumed: fence the floor and the wings — the belly gets a certificate.

Two classical conditions keep a smile arbitrage-free: the **Durrleman function g(k) ≥ 0** (non-negative density — no butterfly) and **Lee's bound** on wing growth. SVI satisfies neither automatically — and a bank-committee review proved the sharpest failure: a fit can pass every parameter fence and still hide **belly** arbitrage (**9.2% of historical fits did**). The answer became a design rule: **an uncertified slice cannot become a mark.**

DURRLEMAN BUTTERFLY FUNCTION + LEE CAP IN JW UNITS

$$g(k) = \left(1 - \frac{k\,w'}{2w}\right)^2 - \frac{w'^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \;\geq\; 0, \qquad b\,(1 + |\rho|) \leq 2$$

g(k) — Durrleman function (g ≥ 0 everywhere ⇔ non-negative density, no butterfly); w, w′, w″ — total variance and its k-derivatives; k — log-moneyness; p, c — JW put/call wing slopes; v — ATM variance; t — expiry; 2 — Lee's cap on the asymptotic variance slope.

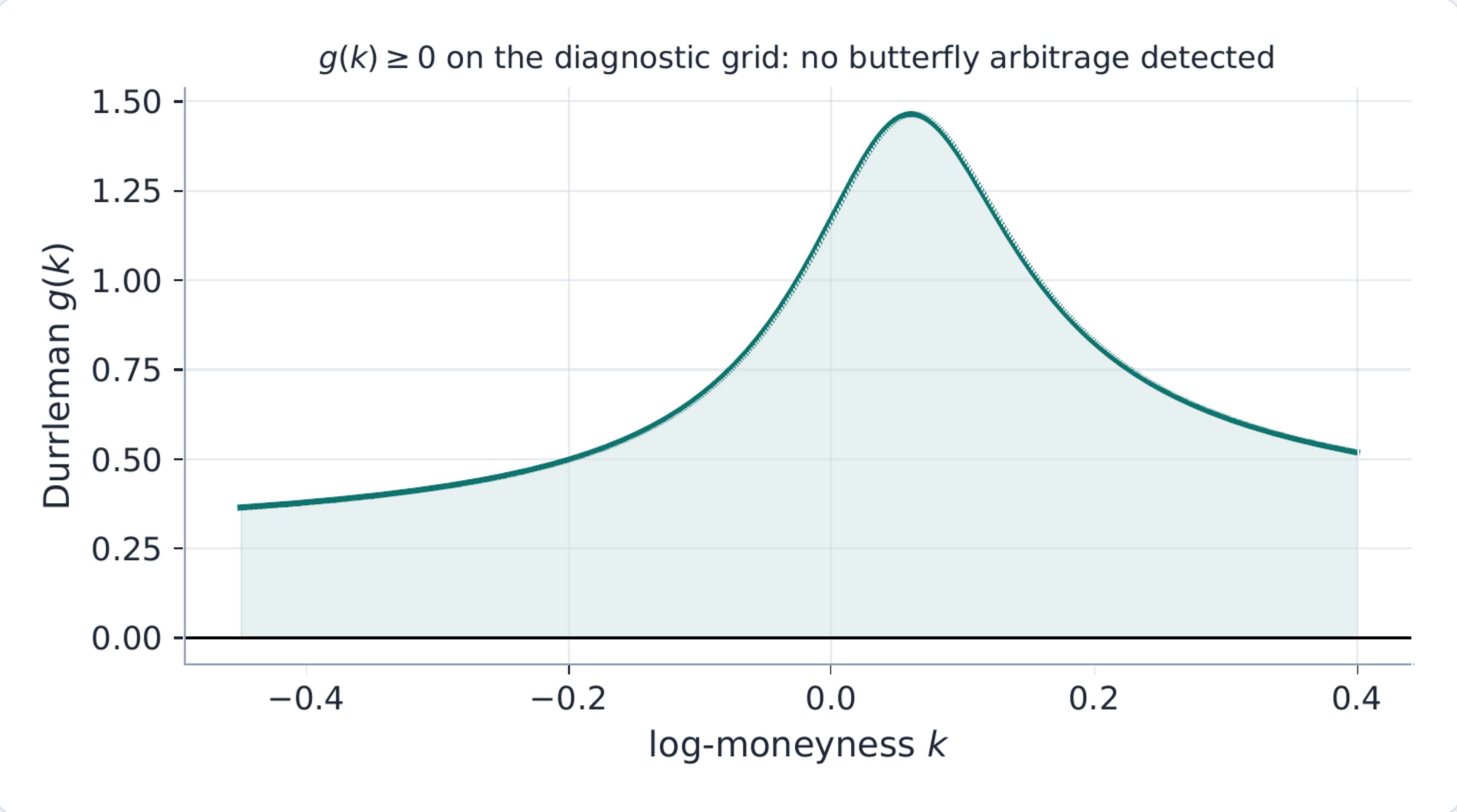
- **The belly certificate:** an 801-point g(k) scan on *every* fit — microseconds — with a one-shot belly repair, kept only if it certifies. Publish is **hard-blocked (HTTP 409)** on an uncertified slice.
- **Fences at a buffered Lee cap 1.95** — at β = 2 exactly a slice passes every screen yet prices negative tail density; the buffer closed a **live trap** (a real SPY LEAP sat on the old wall).
- Even the **measurement** was audited: a finite-difference metric flagged **28.3% of provably-clean LQD slices**; the shipped metric evaluates g(k) analytically — true rate 0.0%.

801-pt

certificate on every fit — publish 409s without it

28.3% → 0.0%

LQD "arb rate" once measured analytically



The diagnostic, plotted. A fitted slice's Durrleman function staying non-negative. The fences bound the wings and the floor; the certificate is the *independent* check — on an arbitrage-free mixture target all three models certify while SVI misses by **118.7 bp** RMS vs LQD **19.7** / MCS **15.0**: the miss is expressiveness, not arbitrage.

TRADING RELEVANCE

A surface you can quote from with eyes open: **LQD is butterfly-free by construction; SVI/MCS are fenced, then certified per fit — and the certificate, not a policy document, is what gates publish**.

Multi-Core Sigmoid, the event-smile model: fitting the W shapes a convex model cannot reach.

Around a scheduled binary event (earnings, CPI, FDA...) the market prices a **mixture of outcomes**, and the smile develops shoulders — a "W". **MCS = Multi-Core Sigmoid**, a kernel-based implied-variance model: a six-parameter sigmoid base that owns the wings, plus up to R localized "hat" kernels (the *cores*) that add the shoulders without being able to touch the tails.

THE SIX-PARAMETER SIGMOID BASE

$$v_{\text{base}}(z) = V_0 + S_0\,(z - z_0) + K_0\,\Phi_\kappa(z - z_0), \qquad \kappa = \kappa_P\ (z < z_0),\ \ \kappa_C\ (z > z_0)$$

V_0 — level; S_0 — slope; $K_0 \geq 0$ — convexity; z_0 — centre; κ_P, κ_C — put/call wing steepness; $z = k/(\sigma_{ref}\sqrt{t})$ — log-moneyness in stdevs; Φ_κ — log-cosh primitive.

THE FULL MODEL — BASE + R SIGNED ZERO-WING HATS

$$v_R(z) \;=\; v_{\text{base}}(z) \;+\; \sum_{r=1}^{R} \alpha_r\, B_{c_r,h_r,\kappa_r}(z), \qquad w(k) = t\, v_R(z)$$

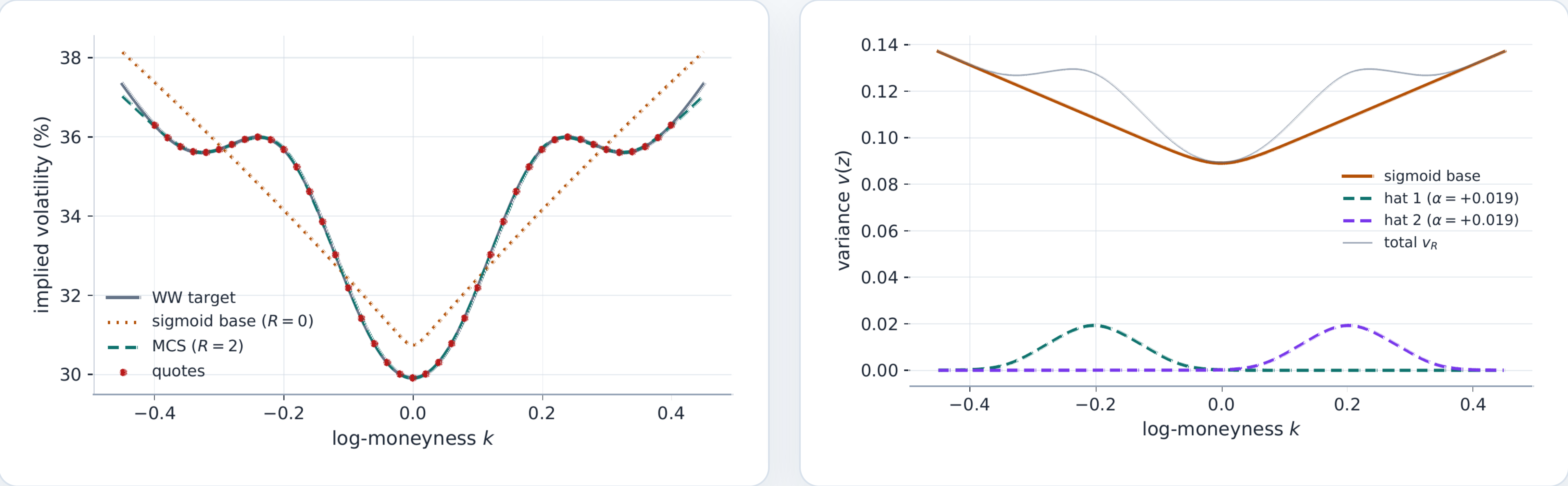
v_R — full MCS variance slice; v_{base} — the base above; R — number of hats (cores); α_r — signed hat amplitude; B — zero-wing hat kernel centred c_r, half-width h_r, steepness κ_r.

THE HAT KERNEL — A NORMALIZED SECOND DIFFERENCE OF THE SAME LOG-COSH PRIMITIVE

$$B_{c,h,\kappa}(z) \;=\; \frac{\Phi_\kappa(u-h) - 2\,\Phi_\kappa(u) + \Phi_\kappa(u+h)}{2\,\Phi_\kappa(h)}, \qquad u \;=\; z - c, \qquad \Phi_\kappa(u) \;=\; \frac{4}{\kappa^2}\log\cosh\!\left(\frac{\kappa u}{2}\right)$$

B — hat kernel: 1 at its centre c, vanishing (with its derivatives) in both wings; h — half-width; κ — steepness; u = z − c.

- **Hats never move the wings** (zero-wing lemma) — analytic Jacobian **3.4× vs FD**; a W the base misses by 153 vol bp fits to **2.55**.



A W-smile, fitted and decomposed. Left — the MCS fit through an event-shaped slice a convex model cannot reach. Right — the decomposition: smooth sigmoid base plus localized hat kernels, each with its own centre, width and sign.

TRADING RELEVANCE

Earnings weeks stop being "the expiry we can't fit" — **the event structure is captured without spline chaos and without contaminating the wings**.

MCS guardrails: a model powerful enough to fit a W is powerful enough to invent one.

Where quotes are dense, the data pins MCS's hats. On a thin chain, the optimizer can park a hat where quotes are stale or missing and "improve" the fit by bending the smile into **negative implied probabilities** — butterfly arbitrage nobody would trade. A census of real illiquid chains found 64% of these violations in the put wing. Three guardrails answer it.

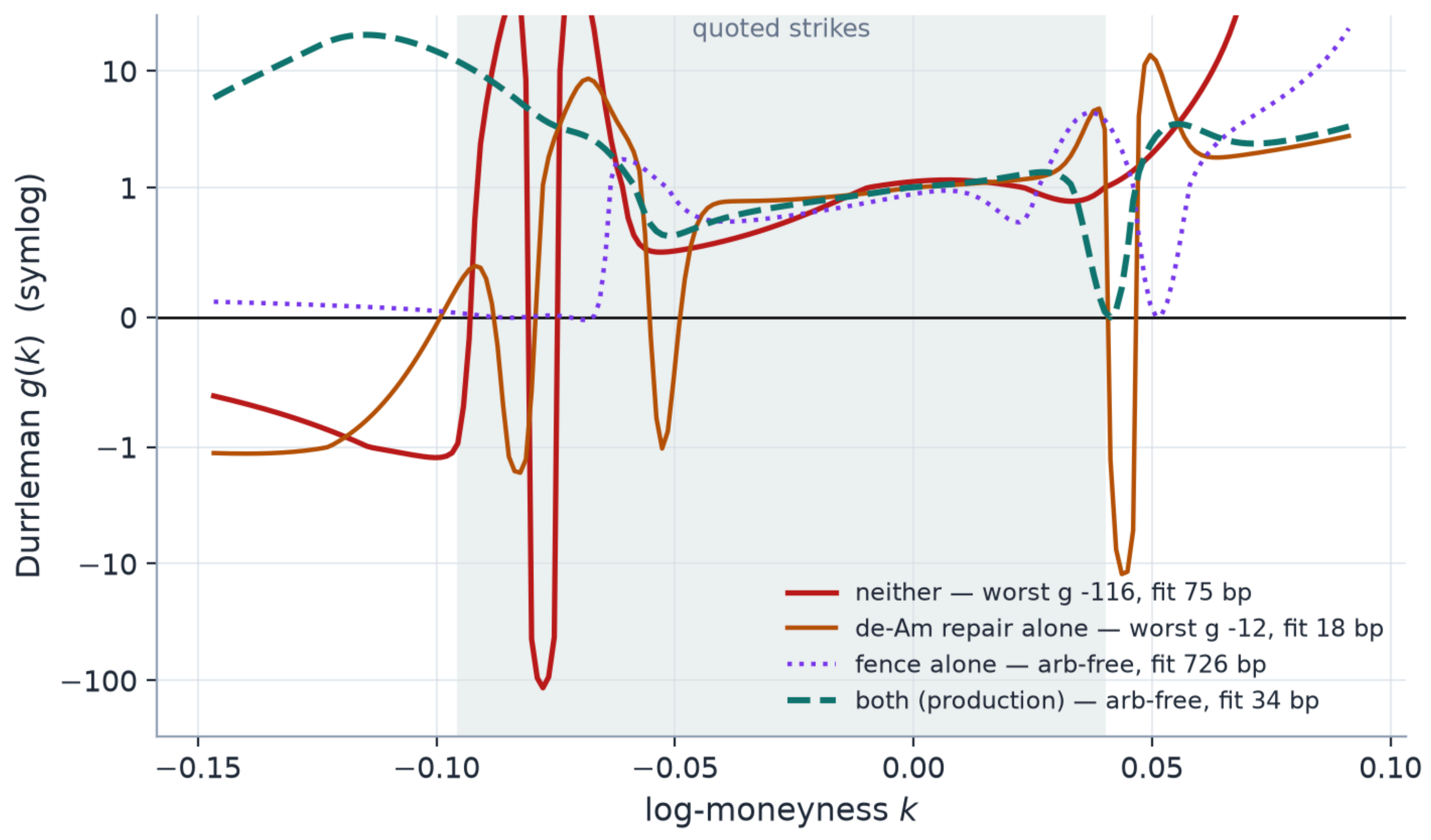
- **1 · A hard budget.** A hat costs 4 parameters on top of the 6 base ones, so hats are admitted only while parameters stay below the quote count — $R \le (N_{quotes}-6)/4$ — and production **caps R at 2 outright**: the backtest showed a third hat buys no out-of-sample accuracy.
- **2 · A fence where the data ends.** The same Durrleman diagnostic g(k) used across models is sampled on a grid extending *past* the quoted strikes, and the fit is charged only where $g < 0$ — twice as hard on the put side, where the census put the violations. The hinge is **exactly zero on a clean fit**: liquid names are byte-identical with the fence on or off.

3 · PROOF BY ABLATION — ONE REAL NODE, ALL FOUR CELLS

Two complementary defences: repairing **arbitraged inputs** (the de-Americanization repair, under Quote preparation) and this **output fence**. The node charted at right — EFA, Aug-2024 vol spike, 11 days out — fitted all four ways:

NEITHER	ONE ALONE	BOTH
75 bp RMS, worst g = −116 — arbitrage and a poor fit	de-Am repair: 18 bp but g = −12 remains · fence: arb-free but **726 bp**	arb-free at **34 bp** — the repair makes the fence affordable

Across all 38 arb-prone census nodes the medians tell the same story: **92 / 25 / 749 / 225 bp**.



The same experiment as the panel — one real chain, four fits. EFA, 22 quotes. No defence (red): g — proportional to the **implied probability density** — dives to −116; the arbitrage lives in the corrupted American put quotes themselves. Repair alone (orange): tight fit, g = −12 remains. Fence alone (violet): arb-free but pays 726 bp fighting the corrupted inputs. Both (teal): arb-free at 34 bp. Grey band — quoted strikes.

TRADING RELEVANCE

Model risk management built in: expressiveness is **budgeted** (parameter caps), **measured** (g(k) reported on every fit), and **ablation-tested on market data** — not left to the optimizer's imagination.

MODELS · LOCAL VOL THE NON-PARAMETRIC PRICING SURFACE

Local Vol, the non-parametric model: a piecewise-affine local-variance surface calibrated through the pricing PDE itself.

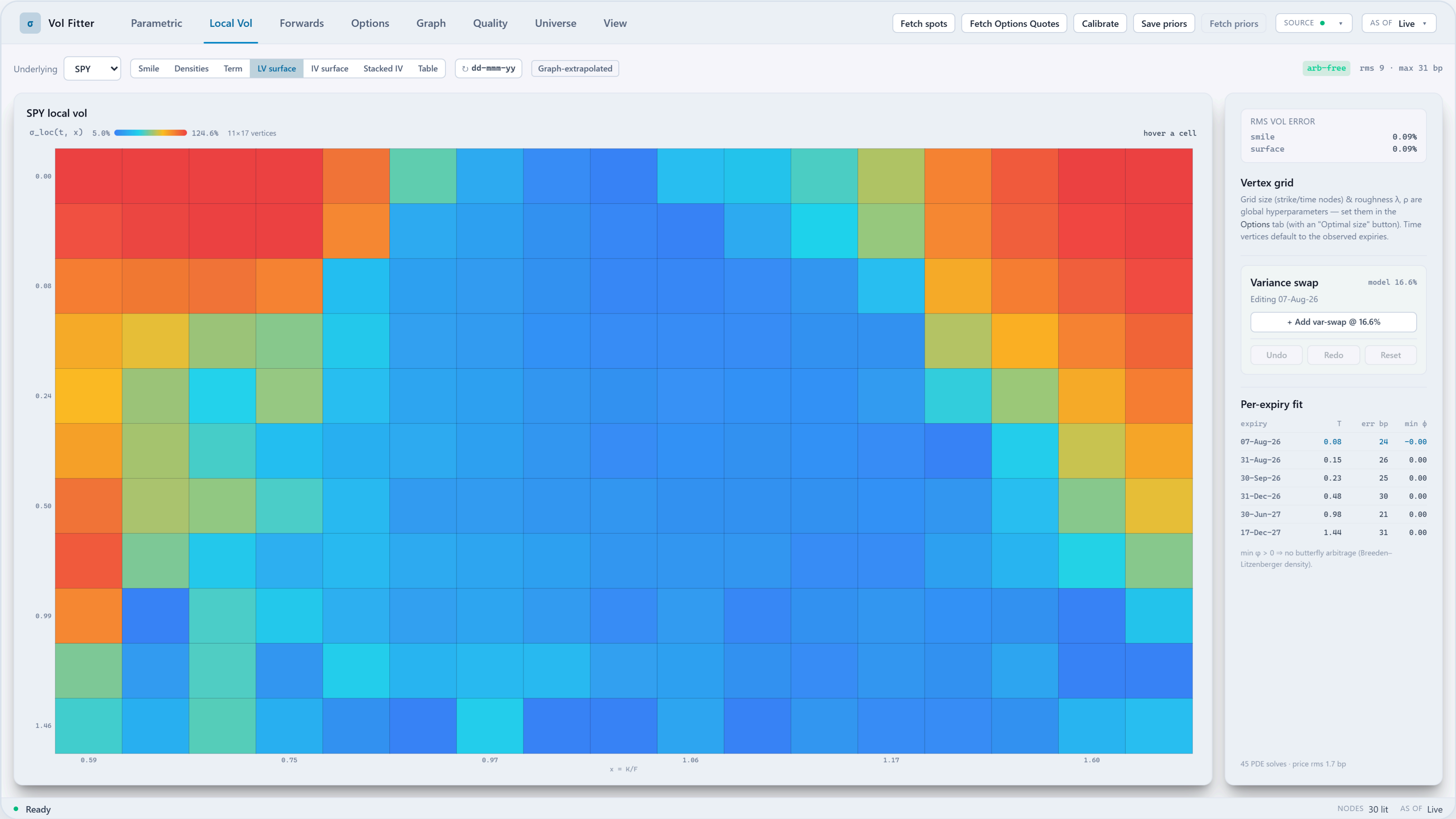
Local volatility is the state-dependent instantaneous volatility σ(t, S) consistent with today's option prices (Dupire). Here it is not extracted from a fitted smile by differentiation; it is **calibrated directly from observed option prices, without modeling the IV curve at all**: the unknowns are local-variance values on a maturity × strike grid, priced through the forward pricing equation (a partial differential equation, PDE) in the spirit of Andreasen–Huge.

FORWARD DUPIRE EQUATION — THE IMPLICIT PRICING MAP

$$\partial_T\, c(T,x) \;=\; \tfrac{1}{2}\,\nu(T,x)\,x^2\,\partial_{xx}c(T,x), \qquad c(0,x) = (1-x)^+$$

$c(T,x)$ — forward-normalized call price; T — maturity (event-weighted variance time); $x = K/F_T$ — normalized strike; $\nu(T,x) = \sigma^2_{loc}$ — the local variance being calibrated; $(1-x)^+$ — payoff initial condition.

- **Piecewise affine, concretely:** each grid rectangle splits into **two triangles**; ν(t,x) is exactly affine per triangle (bilinear would not be), continuous across edges. Every surface value is a convex mix of three nodal variances: **positive nodes ⇒ positive surface**; butterfly and calendar cleanliness of the priced output is **measured per fit**, not assumed.
- Quotes are fitted **through the forward PDE** (Andreasen–Huge philosophy): the surface that prices the market is found directly, not bootstrapped slice by slice.
- **Wings:** beyond the last strike node the local variance extrapolates flat; an opt-in **convexity hinge acts only in the extrapolation tail** (≤ 5Δ put, below the deepest quote) — never on quoted strikes.
- **Wing behaviour is a stated contract** — bounded positive local variance keeps implied wings inside Lee's cap on the grid; cleanliness is checked on the price grid, per regime.



Local Vol workspace — live SPY (Massive feed). The calibrated nodal local-vol heatmap over maturity × strike, with the arbitrage badge, per-expiry error table and var-swap panel. Smile, densities, term structure and implied-vol surface views are all

TRADING RELEVANCE

The surface that reprices the vanilla market drives PDE pricing and the Local Vol views directly; the exact sticky-local-vol scenario runs off a separately extracted parametric-Dupire grid — two stated objects, not one pretending.

MODELS · LOCAL VOL ENGINEERING FOR REAL CALIBRATION LOOPS

The Local Vol calibrator is engineered like a production solver, not a research script.

Calibrating local vol solves the pricing PDE hundreds of times inside an optimizer, so three numerics must be beyond suspicion: a time-stepping scheme that cannot oscillate, exact gradients, and an inner solve fast enough to live in the loop. Each is addressed explicitly.

IMPLICIT EULER STEP — UNCONDITIONALLY STABLE AND MONOTONE

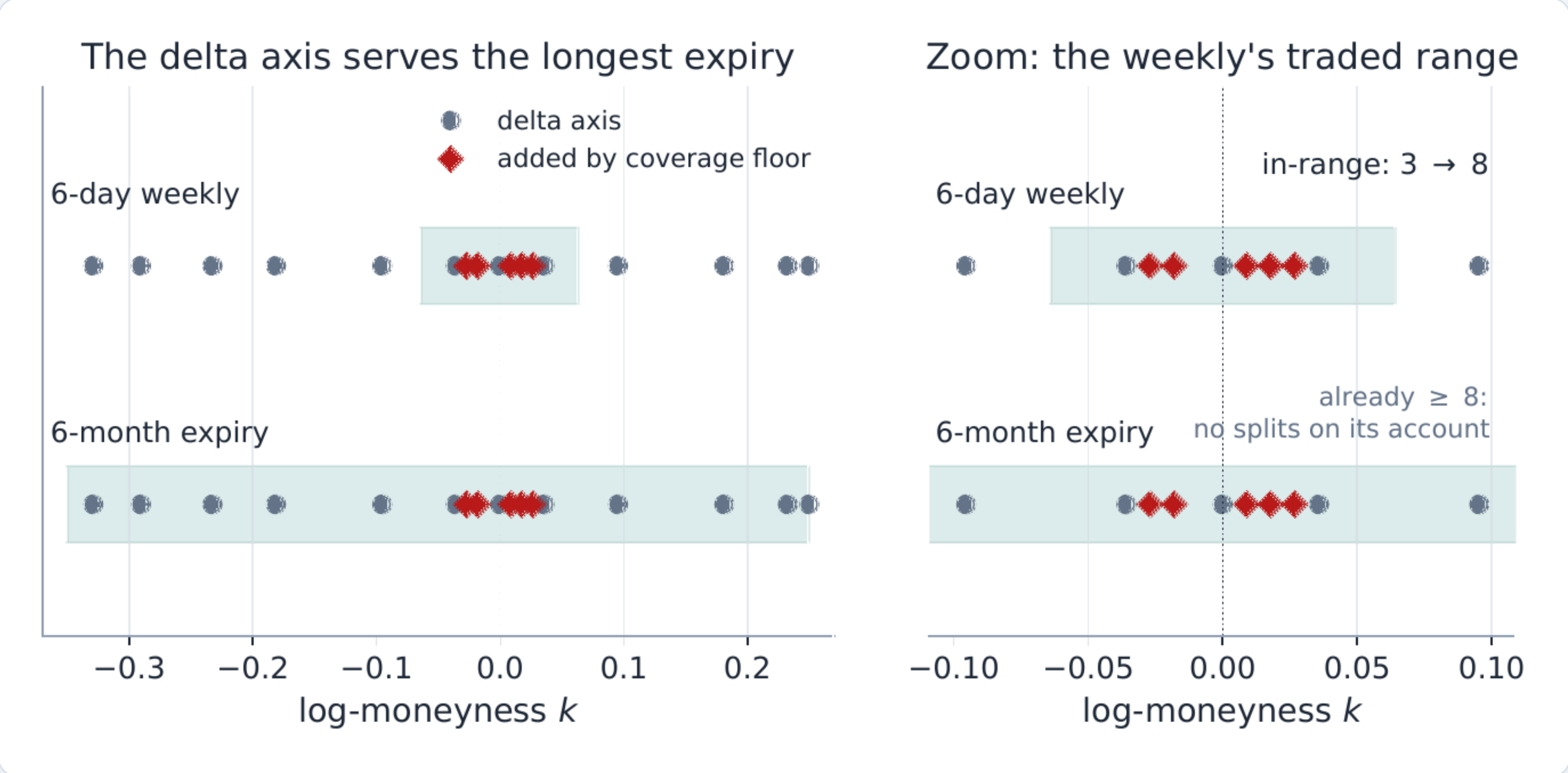
$$B^{n+1} = I - \Delta T\, A^{n+1}, \qquad B^{n+1}U^{n+1} = U^n + \text{boundary}, \qquad B^{n+1} \text{ an M-matrix}$$

U^n — grid call prices at time level n; B = I − ΔT·A — implicit-step matrix (an M-matrix: $B^{-1} \geq 0$ ⇒ ordered, positive prices); ΔT — time step; A — tridiagonal discretization of $\frac{1}{2}\nu x^2\partial_{xx}$ (ν local variance, x normalized strike).

- **Why implicit Euler and not Crank–Nicolson?** CN is more accurate per step but only *monotone* under a step bound — beyond it the payoff kink excites oscillations that read as negative densities. The implicit step is monotone at *any* step ($B^{-1} \geq 0$: prices stay ordered, positive), one order of accuracy traded for the guarantee.
- **Exact tangent gradients of the scheme actually solved** — the march itself is differentiated (every sensitivity reuses the price solve's factorization), FD-checked; a matrix-free solver adds ~1.3–1.65× where it engages.
- **Compiled inner loop (Numba): ~6×** the LAPACK banded solve, identical to ~1e-15, NumPy fallback — plus a stall early-stop worth another 1.5–3× on cold fits.

~6×
PDE march (Numba vs LAPACK)

108 → 23.5 bp
six-day weekly rescued by grid diagnosis



Case file — the weekly that broke the grid. A true 1-week expiry fitted terribly until per-expiry diagnostics (vertices-in-range, vega floor, PDE steps) isolated strike under-resolution as the cause. The fix is measured, not guessed: 108 → 23.5 bp.

TRADING RELEVANCE

Local vol you can recalibrate **inside a trading loop** — stable numerics, exact tangent gradients (bump-and-reprice-checked), and offline per-expiry diagnostics that name the failure mode when a fit degrades.

American quotes are converted to European-equivalent vols before any model sees them.

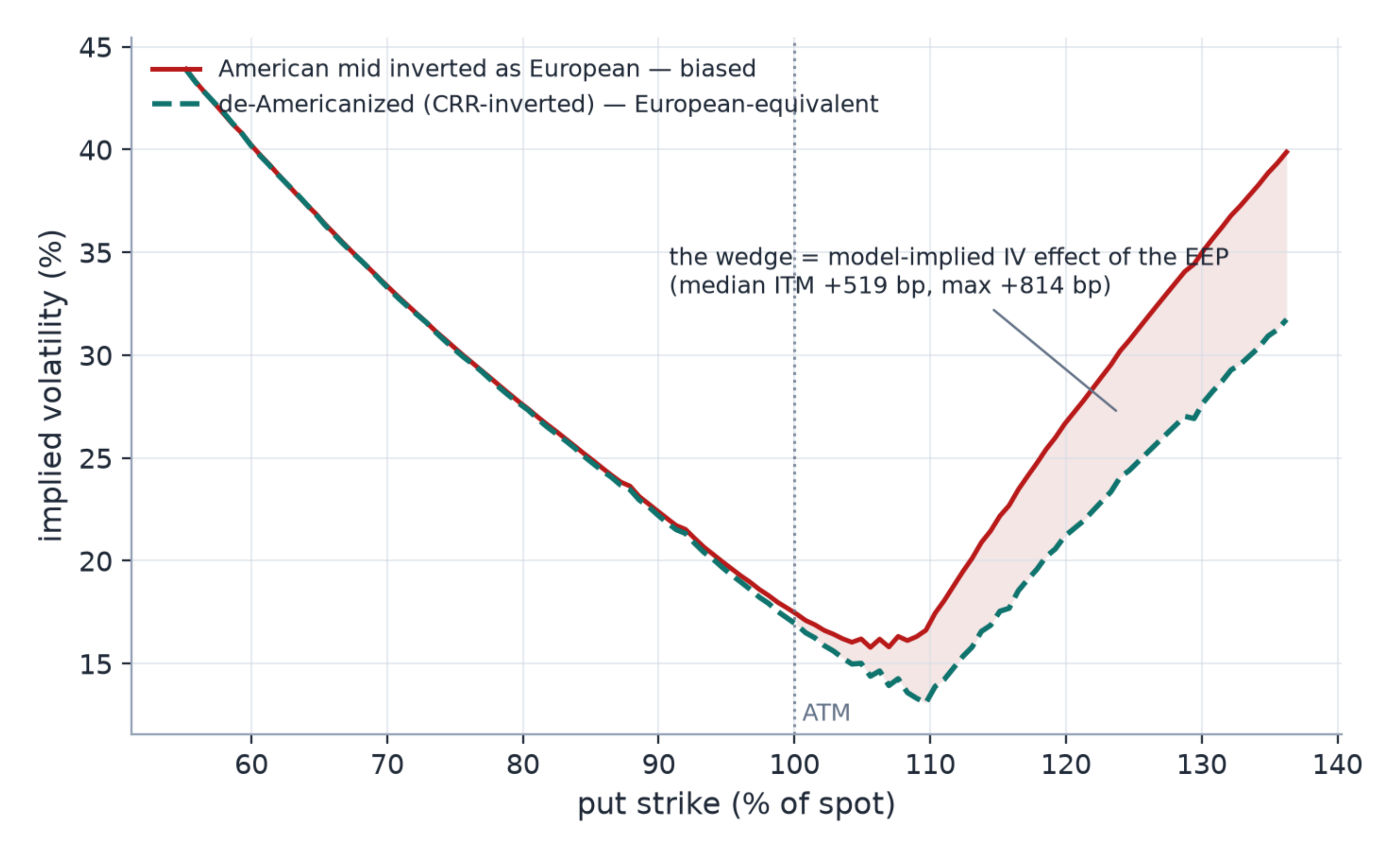
US-listed single-stock and ETF options are **American**: they can be exercised early, so their quotes embed an early-exercise premium a European model does not know about. A **CRR binomial tree** values that right (the exercise decision rolls backward through a discrete spot lattice), and each quote is inverted through it: find the vol at which the American tree price matches the quote — the premium is removed, strike by strike.

PER-STRIKE CRR INVERSION — RETURN THE EUROPEAN-EQUIVALENT BLACK VOL

$$\text{find } \sigma^* \text{ s.t. } \mathrm{CRR}_{\mathrm{Am}}(\sigma^*) = A^{\mathrm{quote}}, \qquad p = \frac{e^{(r-q)\Delta t} - d}{u - d}$$

A — observed American option price; $\mathrm{CRR}_{\mathrm{Am}}(\sigma)$ — American value from the Cox–Ross–Rubinstein binomial tree at vol σ (given spot, strike, expiry, rate, dividends); σ* — the root, returned as the European-equivalent Black vol.

- **The tree only prices the American/European difference** — discretization error largely cancels in that difference (the control-variate role). Discrete cash dividends use an **escrowed-spot lattice**: the tree diffuses spot net of future-dividend PV (cash drops would stop it recombining), added back per node so exercise is checked against the true cum-dividend spot.
- **Compiled CRR kernel (Numba): ~3 ms where NumPy takes ~350 ms** on a ~300-quote chain — cached under a digest of the only inputs de-Am depends on: re-tuning a fit reuses the conversion; trees re-run only on a data change.
- **Caution, learned in production:** a whole-curve convexity repair gapped live SPY/NVDA smiles — its free *global* baseline let a wing repair re-tilt the curve, moving the dense, heavily-weighted ATM core a sub-penny: a visible discontinuity. Reverted; the shipped repair is **wing-only, band-constrained**, ATM core exactly fixed.



Real market data, not a toy. One live SPY expiry (Massive capture, Jun 2026, ~6 months): every put mid inverted two ways — as if European (red) vs de-Americanized through the tree (teal). Out of the money the two agree to a few bp; in the money the wedge is the model-implied IV effect of the early-exercise premium — median **+519 bp**, up to **+814 bp**. Honestly labelled: production fits only the OTM side, so these ITM puts are the discarded twins of fitted OTM calls; the population actually fitted sees a median ~4 bp, up to ~60 bp in the put wing — still material against sub-bp fit budgets.

TRADING RELEVANCE

Single names and dividend payers become fittable: **early exercise is removed before modeling, and the repair never invents information where the market quoted none**.

QUOTE PREPARATION FORWARDS & DIVIDENDS · CASE FILE INSIDE

Forwards are implied from the chain itself — and the worst-identified parameter is clamped, not trusted.

Every implied vol stands on a forward F and a discount D. Both are estimated from the chain itself — put–call parity holds model-free, so regressing C−P on strike recovers them with no external curve. The catch: the **slope** (the discount) is poorly identified on noisy chains, and an error there drags the forward and tears the ATM smile.

PUT–CALL PARITY AS A REGRESSION — SLOPE IS THE DISCOUNT, LEVEL IS THE FORWARD

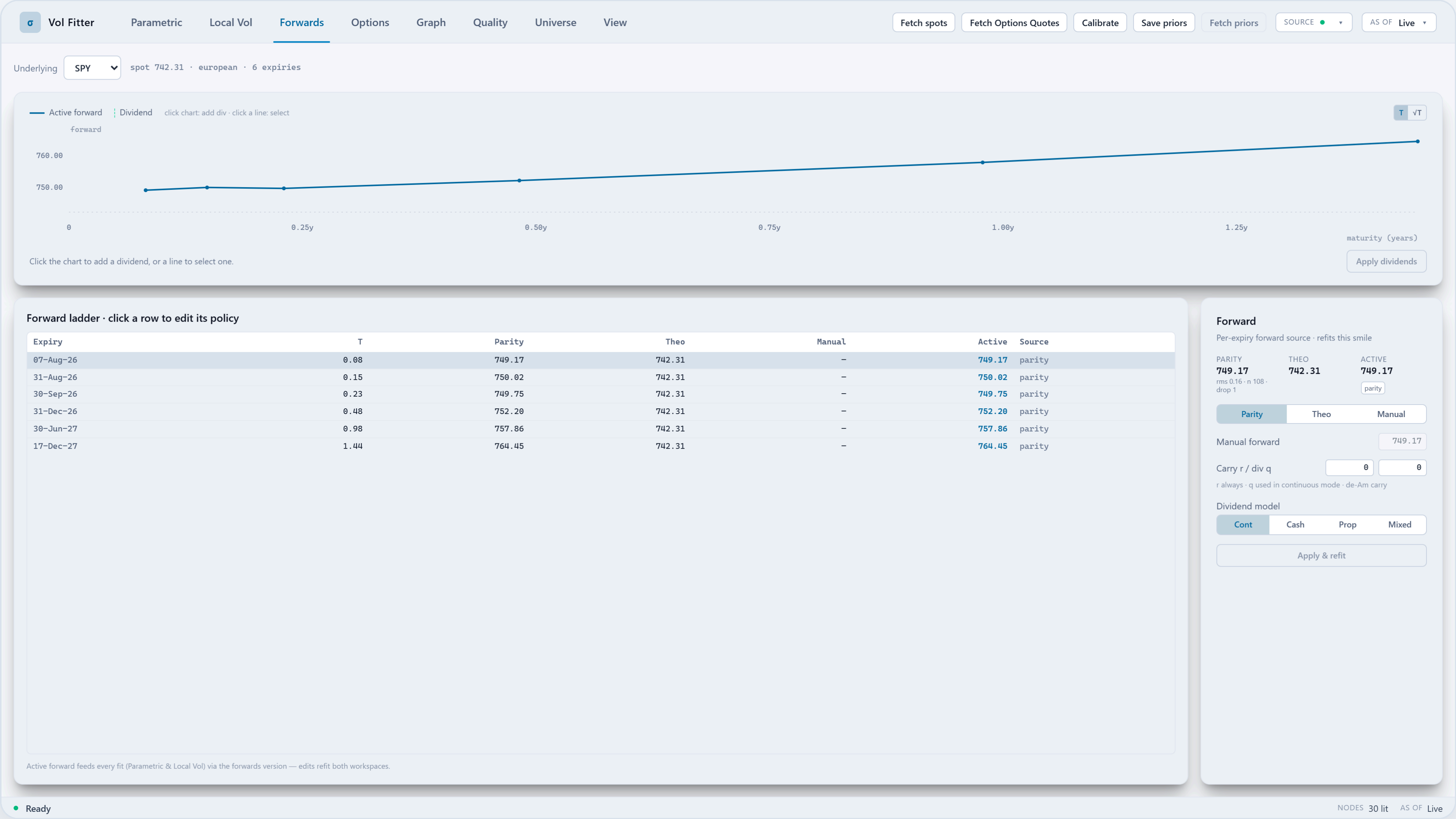
$$\underbrace{C(K) - P(K)}_{y} = \underbrace{D\,F}_{a} - \underbrace{D}_{-b}\,K \quad\Longrightarrow\quad D = -b, \quad F = a/D$$

C(K), P(K) — call/put mids; D = e^{-rT} — discount (= minus the slope); F — forward (= intercept / discount); a, b — the least-squares line C−P = a + bK.

CASE FILE — THE DELAYED FEED THAT GAPPED THE ATM SMILE

A delayed tier quoted stale pairs; the parity **slope** drifted to discounts above 1 (negative rates), dragging the forward and tearing the ATM smile. Fix: **clamp the implied rate to a physical band**, re-derive the forward from the well-identified level. Clean chains byte-identical; the broken feed heals.

- Four dividend modes; a de-bias loop removes American bias with the discount held fixed.
- **Zero-carry synthesized chains carry no parity signal** — flagged at ingestion, pinned to F = spot, D = 1 (badged), never regressed.
- **Borrow, solved jointly (opt-in):** a de-Am ↔ parity fixed point (2–4 iterations) with closed-form IV-vs-borrow sensitivity; thin boards read **"unidentified"**, never an invented borrow.



Forwards workspace — live SPY (Massive feed). The forward curve with per-expiry parity-implied vs theoretical carry values, the source of each, manual overrides, and a click-to-edit discrete dividend schedule that feeds both the forward model and the de-Americanization lattice.

TRADING RELEVANCE

Every IV in the system stands on a **defensible forward** — estimated from the market, stabilized against feed pathologies, and overridable when the desk knows better.

CALIBRATION ONE OBJECTIVE, EVERY MODEL

The calibration objective: vol error, inside the quoted spread — with three fit targets (mid, band, haircut).

A mid quote is not fair value — it is the midpoint of a spread that can be half a vol point wide in the wings. So all models share one objective: residuals normalized by **vega** (the loss reads in vol units), **bid–ask bands** (only leaving the quoted band costs anything), and **liquidity-aware weights**.

VEGA-NORMALIZED PRICE RESIDUAL — BEHAVES LIKE VOL ERROR, ITERATES STAY VALID PRICES

$$r_i \;=\; \frac{C_i^{\mathrm{mod}} - \mathrm{Bl}(k_i, w_i)}{\partial_\sigma \mathrm{Bl}(k_i, w_i) + \eta}$$

r — residual; $C^{\mathrm{model}}(k)$ — model call at log-moneyness k; B(k,w) — Black price at the quote's total variance w; $\partial_\sigma B = \varphi(d_+)\sqrt{T}$ — Black vega (φ — standard normal pdf); η — small vega floor for the far wings; T — expiry.

BID–ASK BAND OBJECTIVE — INSIDE THE BAND, ONLY A TINY MID ANCHOR

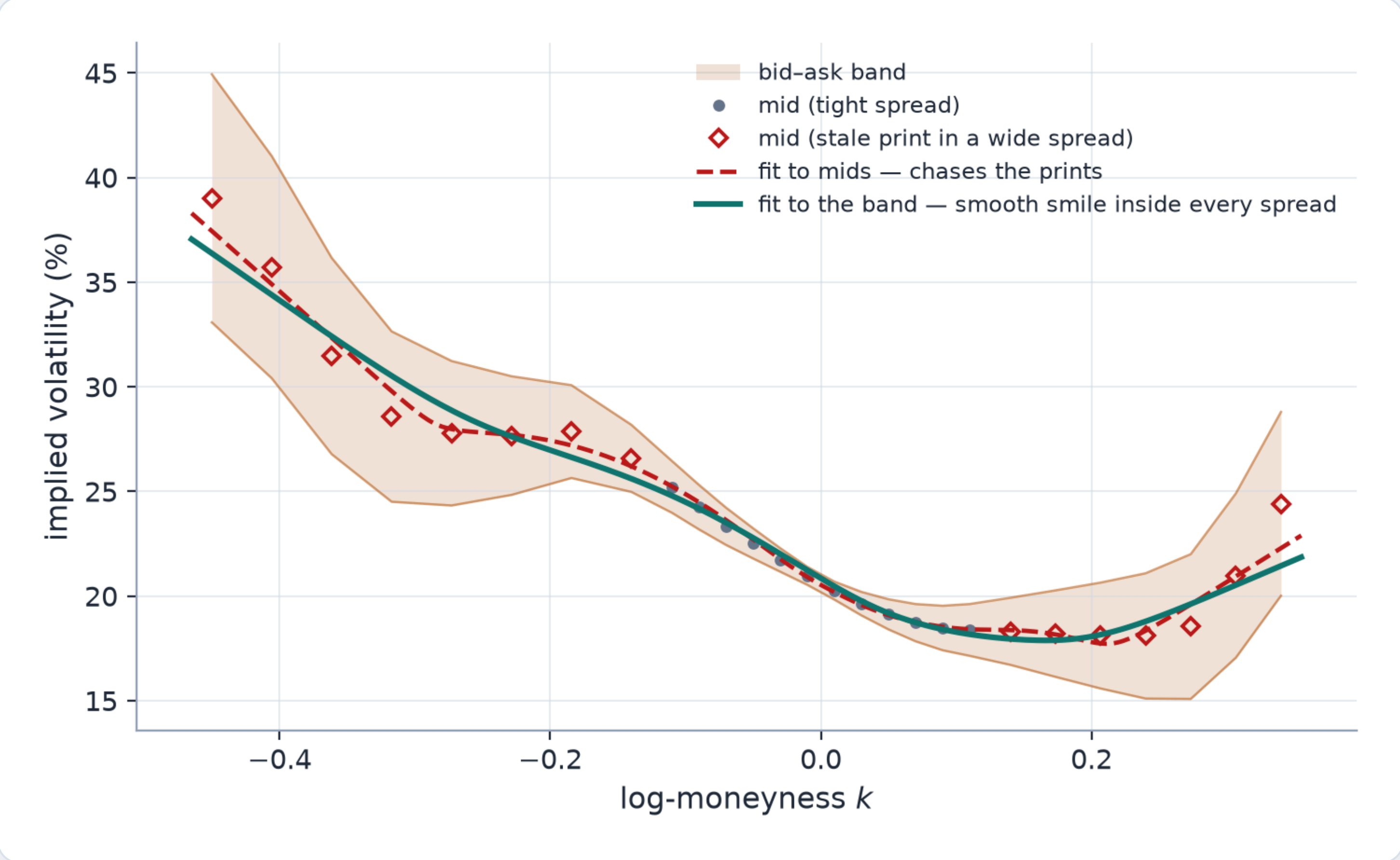
$$\mathrm{loss}_i = \max(m_i - \mathrm{hi}_i, 0)^2 + \max(\mathrm{lo}_i - m_i, 0)^2 + \theta_{\mathrm{mid}}\,(m_i - \mathrm{mid}_i)^2$$

m_i — model value for quote i; lo_i, hi_i — bid–ask band edges (haircut-shrunk in haircut mode); mid_i — mid quote; θ_{mid} — small mid-anchor weight (0.05); inside the band only the anchor term is paid.

DENSITY-CORRECTED TIME-VALUE WEIGHTS — STRIKE COVERAGE, NOT QUOTE COUNT

$$\omega_i \;=\; \max(\mathrm{TV}_i, \varepsilon)\cdot\frac{s_i}{\bar{s}} \qquad (s_i = \text{Voronoi cell width in } \log K)$$

ω_i — quote weight; TV_i — time value (the OTM price in forward units — an OTM option's whole price is time value), so richly-priced strikes carry more weight; ε — tiny numerical floor; s_i — the quote's 1-D **Voronoi cell** width: the stretch of strike axis closer to this quote than to any other (half the gap to each neighbour), i.e. how much strike range this quote alone covers. Multiplying by $s_i/\bar{s}$ ($\bar{s}$ — mean spacing) divides out the exchange's listing density: without it, the densely-listed ATM cluster would outvote the wings simply by quote `count`. On a uniform strike grid the correction is exactly 1.



Same model, same settings — only the target differs. Noisy wing mids (stale prints inside wide spreads): the **fit to mids** (dashed) chases them into kinks; the **fit to the band** (teal) stays a smooth smile inside every quoted spread.

- **The haircut target** shrinks every band by a fixed vol margin: tight ATM spreads collapse onto the mid, wide wing spreads keep a meaningful band — one dial spans mid- and band-fitting.
- **Every excluded quote is named** — a seven-reason quarantine + a 3-tick deep-OTM floor on the *bid*; the drop list ships with the export.

TRADING RELEVANCE

A fit inside the spread is **free by design** — the fitter only pays to leave the quoted band, which is exactly how a trader thinks about it.

CALIBRATION VARIANCE-SWAP TARGETS · CASE FILE INSIDE

One var-swap quote pins the whole wing mass — entered as a soft target shared by every model.

A **variance swap** pays realized variance; its fair strike is a model-free integral over the whole smile, dominated by the wings. Desks often trust a dealer var-swap mark more than any individual deep-wing quote — so the app accepts one var-swap level per (name, expiry) and lets it discipline the wings of whichever model is fitting.

LOG-CONTRACT REPLICATION + THE PENALTY RESIDUAL

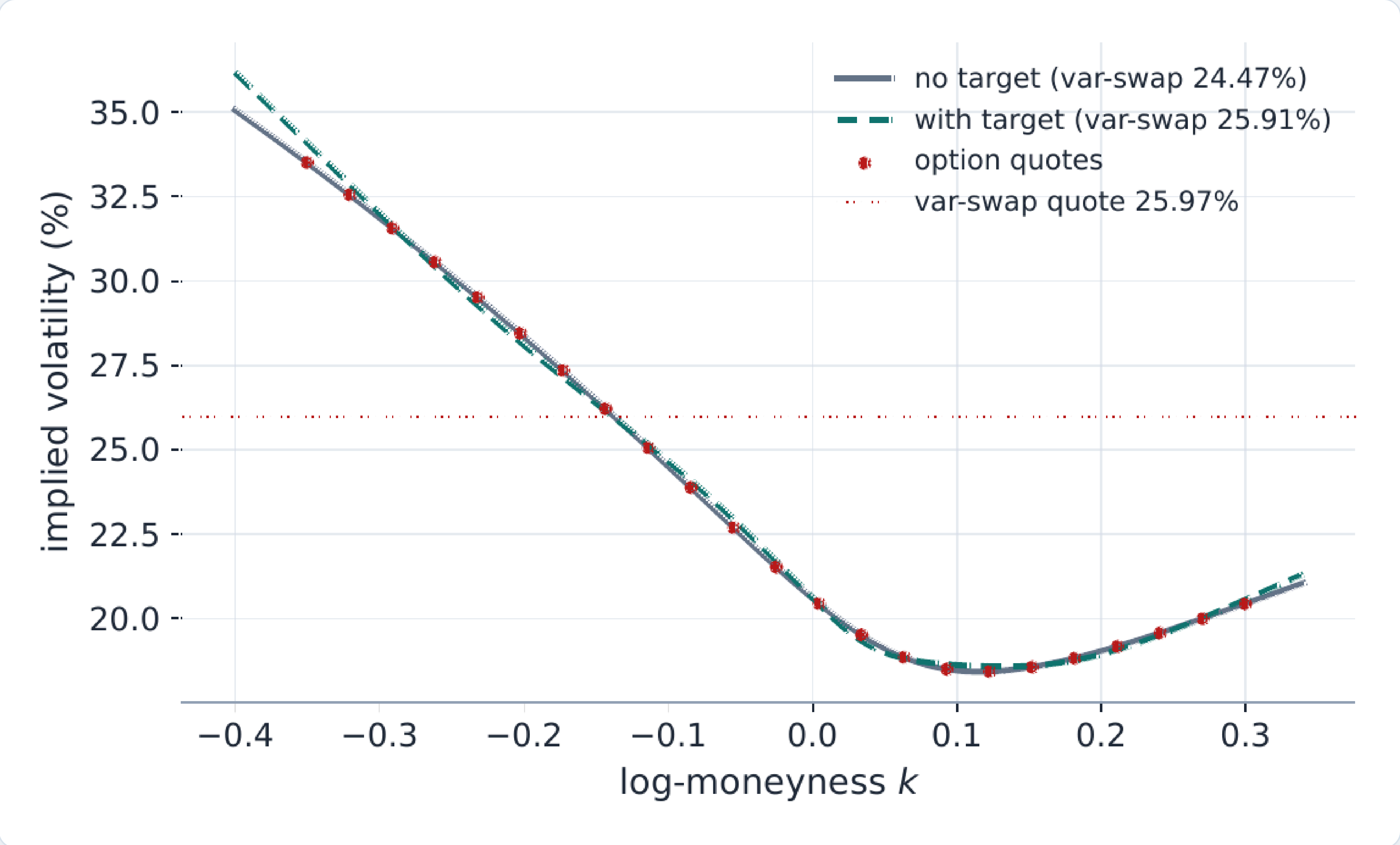
$$w_{\mathrm{vs}}(T) \;=\; 2\int_0^F \frac{P(T,K)}{K^2}\,\mathrm{d}K + 2\int_F^\infty \frac{C(T,K)}{K^2}\,\mathrm{d}K, \qquad r_{\mathrm{vs}} \;=\; \sqrt{\omega_{\mathrm{vs}}}\,(\sigma_{\mathrm{vs}}^{\mathrm{model}} - \sigma_{\mathrm{vs}}^{\mathrm{quote}})$$

w_{vs} — fair var-swap total variance; B(k,w(k)) — Black call on the smile w(k); k — log-moneyness; $(e^k-1)^-$ — put-leg (negative-part) term; r_{vs} — penalty residual; ω_{vs} — weight (a % of the node's option weight); $\sigma_{vs} = \sqrt{(w_{vs}/T)}$ — var-swap vol; T — expiry.

● CASE FILE — THE FIT THAT TOOK MINUTES

Routing the penalty through generic replication meant **~800 Black inversions per Jacobian column** for LQD — a 158-second fit. The lesson became a design rule: **route through the model's native object**. LQD prices the var-swap in closed form, SVI/MCS by arithmetic replication, Local Vol by one extra source-PDE solve.

- Per-node quote with its own **undo/redo session**, shared by Parametric and Local Vol; weight set as a % of total quote weight; byte-identical when off.
- The var-swap penalty now rides the **analytic-Jacobian path** (closed-form derivative for LQD) — the target no longer costs the fit its speed.



The target at work. A fitted smile with and without the var-swap penalty — one number pulls the integrated wing mass to the quoted level without distorting the well-quoted body.

TRADING RELEVANCE

Var-swap and OTC marks become **calibration inputs, not afterthoughts** — the wings integrate to the level the desk actually trades.

CALIBRATION WING & TAIL DISCIPLINE

Wing extrapolation obeys moment logic — Lee's bound caps every model's tails, each by its own mechanism.

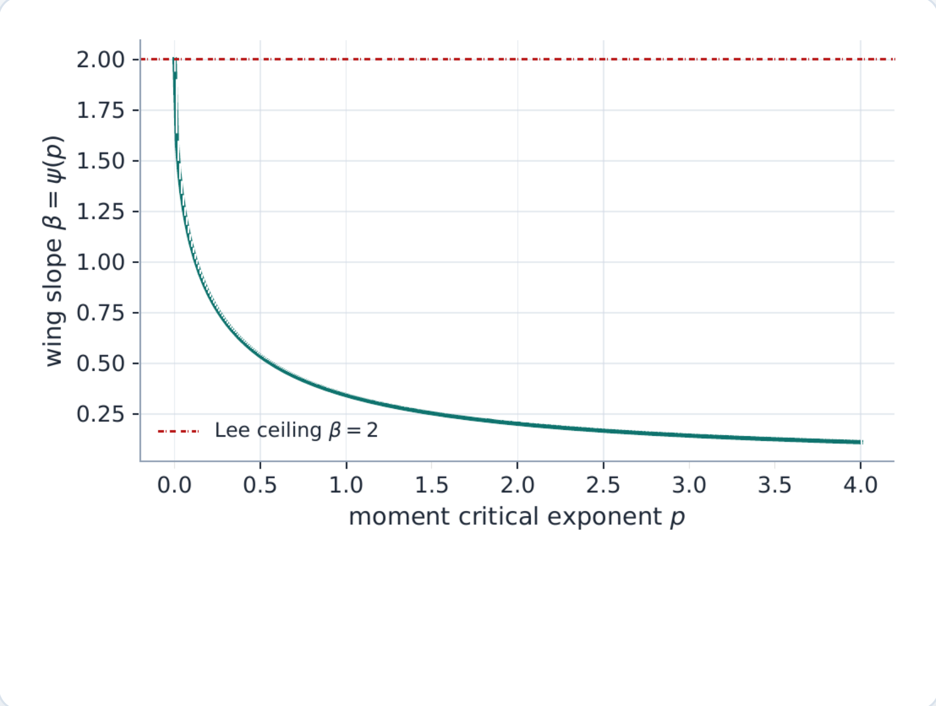
Lee's moment formula: total implied variance can grow at most linearly in |log-moneyness|, with slope at most 2, and the slope is tied to how many moments the terminal distribution has. A steeper wing implies an infinite price for some traded payoff — free money. It is the one universal law of smile extrapolation.

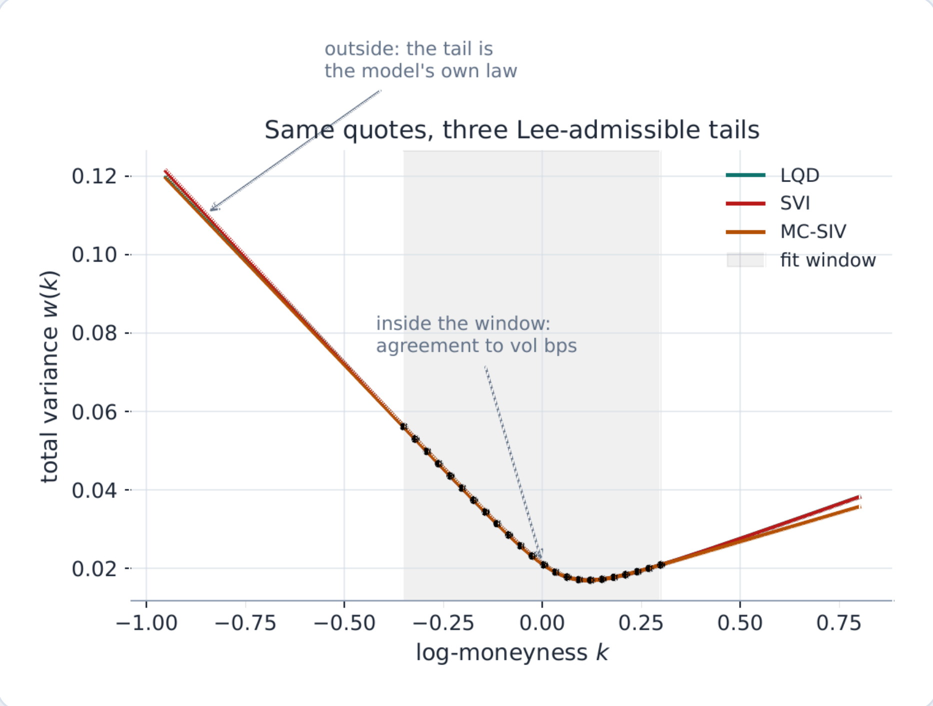
LEE'S MOMENT FORMULA — ASYMPTOTIC VARIANCE SLOPES CAPPED AT 2

$$A_R < 1, \qquad \beta_L = \psi(1/A_L), \quad \beta_R = \psi(1/A_R - 1), \qquad \psi(p) = 2 - 4\big(\sqrt{p^2+p} - p\big)$$

β_L, β_R — left/right asymptotic slopes of total variance $w(k)/|k|$; ψ — Lee's moment function $\psi(p) = 2-4(\sqrt{p^2+p}-p)$; p — critical moment exponent of S_T/F_T; A_L, A_R — LQD endpoint scales (quantile tail slopes).

- **LQD:** structural — the endpoint scales A_L, A_R imply the slopes; admissibility is one inequality.
- **SVI:** a strict fence at a **buffered cap 1.95** — at Lee's bound exactly (β = 2) a slice can pass every screen yet price negative tail density, so the cap sits strictly inside; the buffer closed a live trap (a real SPY LEAP sat exactly on the old wall at 2.0).
- **MCS:** hats are wing-neutral by construction — the base alone owns the tails; the base slopes' Lee admissibility is *diagnosed*, not enforced.
- The far wing is where models **extrapolate with confidence they don't have** — so extrapolation behaviour is a stated, tested contract per model.





Tails, quantified. Left — Lee's $\psi(p)$: the exact map from the number of finite moments to the admissible variance slope. Right — how each model extrapolates beyond the last quote. The curves deliberately disagree: past the data, a wing is a modelling *choice*, not an estimate — what matters is that each model's Lee status is stated and checked: structural for LQD, soft-capped for SVI, bounded for LV, diagnosed for the MCS base.

TRADING RELEVANCE

Deep OTM marks inherit **tail discipline from moment theory** — not from whatever the last two quotes happened to say.

Calendar-arbitrage constraints act only where data lives — a production incident made that a design rule.

Calendar arbitrage: if an option with more time is cheaper than one with less (same forward-moneyness), a calendar spread earns free convexity. The clean statement is in total variance $w = \sigma^2 T$, which must be non-decreasing in maturity. The subtlety is not the constraint — it is **where you are allowed to check it**.

CALENDAR CONSISTENCY — CONVEX ORDER, FOUR EQUIVALENT FACES

$$T_i < T_j : \quad C_{T_i}(k) \le C_{T_j}(k) \iff w_{T_i}(k) \le w_{T_j}(k)$$

$T_i < T_j$ — near/far expiries; $C_T(k)$ — normalized call; $Y_T = S_T/F_T$ — forward-normalized underlying; $\le_{cx}$ — increasing convex order; $G_T(\alpha)$ — integrated upper-quantile curve of Y_T; $w_T(k)$ — total implied variance; k — log-moneyness.

THE SOFT FLOOR HINGE — CONFINED TO THE TRADED STRIKE RANGE

$$r_{\mathrm{cal}} = \sqrt{W_{\mathrm{cal}}}\,\max\big(w_{\mathrm{near}}(k_m) - w_{\mathrm{far}}(k_m),\, 0\big), \qquad k_m \in \text{traded range only}$$

w_{near}, w_{far} — total-variance curves of the nearer/farther expiry; k_m — constraint grid points, confined to the traded log-moneyness range; W_{cal} — calendar hinge weight; the hinge is nonzero only where the near slice exceeds the far.

THE CONFINEMENT PRINCIPLE

A penalty that compares **two curves** may only be sampled where **both are pinned by data** — extrapolated regions can manufacture violations that do not exist.

CASE FILE — THE PHANTOM CALENDAR ARBITRAGE

Far-dated **NVDA and SPY fits came back flattened**. Root cause: the calendar floor was sampled on a fixed wide strike grid, where the near expiry's **extrapolated** steep wing towered over the far expiry — a violation that existed only in the extrapolation, never in the market.

SYMPTOM	DIAGNOSIS	FIX
Far smiles flattened; **151 bp** of phantom penalty in the fit	Floor grid extended into unquoted strikes; wings compared where **neither slice had data**	Floor confined to the **traded log-strike range** → phantom penalty **151 → 0 bp**, real violations still caught

- The hinge is **exactly zero on admissible surfaces** — calendar enforcement is byte-identical to OFF when the surface is clean; the same machinery is model-agnostic.
- The default surface solver is **symmetric**: every expiry fits independently, interfaces are screened for *identified* violations on common quote support, and only violation-connected runs are jointly repaired — no traversal-order bias; clean ladders are exactly their independent fits (a far fit read 10.6 bp free, **1,095 bp** under a naive full-grid floor, 10.6 bp confined — now a certification case).

TRADING RELEVANCE

No-arbitrage enforcement that **never distorts a clean market**: violations are repaired where quoted, and never *invented* from extrapolation. The extrapolated region has its own opt-in machinery — tapered enforcement in the fit and a publish-time wing projection (Notes 09–10).

CALIBRATION THE EVENT VARIANCE CLOCK

Events are modeled as time, not as smile damage — the variance clock is price-preserving by construction.

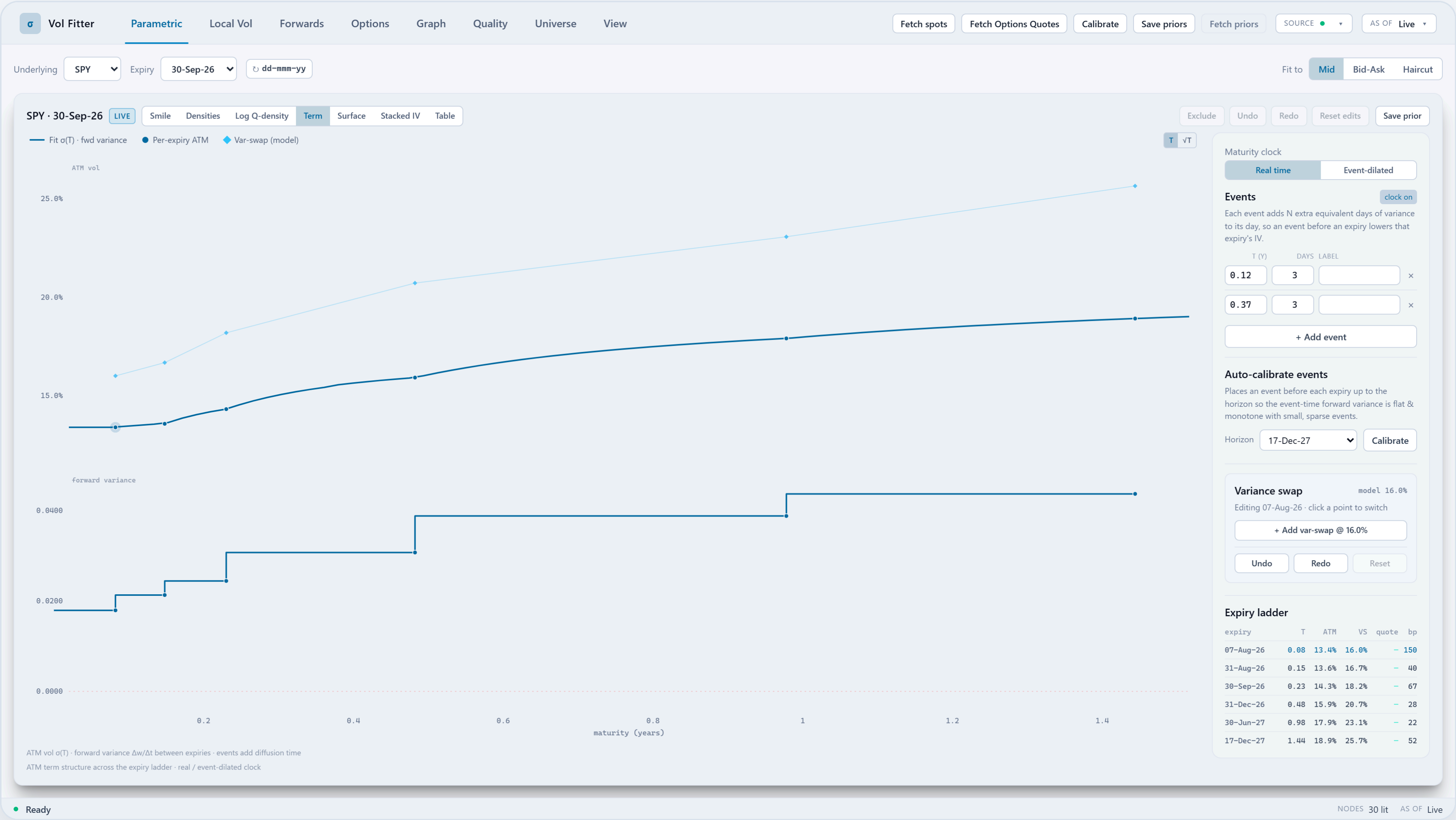
An earnings date concentrates a known lump of variance into a single day — which is why implied vol "crushes" after the print. Modeled correctly, this is a **change of clock**: measure maturity in variance time τ, where an event day counts as 1 + N ordinary days, and the kink in the term structure disappears.

THE DILATED CLOCK — EACH EVENT ADDS N EQUIVALENT DAYS OF VARIANCE

$$\tau_{\text{days}}(t) \;=\; 365\,t \;+\; \sum_{t_e \le t} N_e, \qquad \sigma_{\text{work}}(k) = \sqrt{w(k)/\tau}$$

t — calendar time to maturity (years); t_e — event date; N_e — extra equivalent days of variance carried by the event; τ(t) — variance time in years (fits and IV = √(w/τ) live on τ; with no events τ = t).

- **Total variance w(k) is clock-independent** — prices never move when an event is inserted; only the working IV drops. The vol crush is a change of units, not of value.
- One clock per node, shared by **every view**: fits, local vol, term structure, tables.
- **Auto-calibrate** places events so event-time forward variance is flat — the calendar is inferred from the term structure, then editable.
- Empty calendar ⇒ τ = t, **byte-identical** — the feature costs nothing when unused.
- **Below one day, the same idea continues:** an opt-in intraday session clock accrues variance through the trading session to the exact settlement instant (NYSE calendar, half-days included) — the 0DTE extension, validated on a 312-snapshot intraday capture.



Term workspace — live SPY (Massive feed) with an event calendar. ATM vol and forward variance across the expiry ladder, two scheduled events (2 equivalent days each) in the editable calendar, the real-time / event-dilated clock toggle, event auto-calibration, and the per-expiry var-swap levels.

TRADING RELEVANCE

Pre/post-earnings surfaces become **comparable across names and dates** — event-time vol is the signal; calendar-time vol is the artifact.

DYNAMICS SPOT–VOL TRANSPORT (SSR)

When spot moves, the surface moves by rule, before recalibration — one transport realizes any skew-stickiness ratio.

SSR = skew-stickiness ratio: how much ATM vol moves when spot moves, in units of the skew. The classic regimes are points on one dial — R = 0 "sticky-moneyness" (the smile rides with spot), R = 1 "sticky-strike" (vol per strike frozen), R = 2 "sticky-local-vol". One analytic transport realizes any value in between, instantly and without refitting.

THE SSR TRANSPORT — ONE FORMULA SPANS STICKY-STRIKE TO STICKY-MONEYNESS AND BEYOND

$$\sigma_{\text{new}}(k) = \sigma_{\text{old}}(k+\delta) + (\text{SSR}-1)\, s_0\, \delta, \qquad \Delta\sigma_{\text{atm}} = \text{SSR}\cdot s_0 \cdot \delta, \qquad \delta = \log \frac{F_{\text{new}}}{F_{\text{old}}}$$

σ_{new}, σ_{old} — post-/pre-move smiles in log-moneyness k (vs the prevailing forward); $\delta = \ln(F_{new}/F_{old})$ — log-forward move; SSR — skew-stickiness ratio (0 sticky-moneyness, 1 sticky-strike, ≈2 sticky-local-vol); s_0 — ATM skew $\partial\sigma/\partial k$ at k = 0.

- **SSR is the dial**: R = 0 sticky-moneyness, R = 1 sticky-strike, R = 2 sticky-local-vol — and anything between, from one analytic map.
- **Sticky-local-vol, exactly:** freeze the local-vol surface in absolute strike and move the forward — the new smile is then fully determined: the old curve read at **Hagan's relabeled log-moneyness** $\ell(k,\delta)$, which ≈ $k + 2\delta$ for small moves — precisely where SSR ≈ 2 comes from.
- The calibrated **LV grid** transports by one bookkeeping rule: its strike nodes follow the fraction 1 − R/2 of the spot move (all of it at R = 0, half at R = 1, frozen in strike at R = 2) — pure node relabeling, no refit; only the fully exact sticky-LV reprice costs one Dupire PDE solve.
- **The classic sign trap is documented, not discovered:** a transported *curve* reads the old smile at $k + \delta$, while a fixed-strike *quote*'s moneyness label moves to $k - \delta$. Both signs are correct and both are in production; mixing them flips the skew response — so they live in a stated assumption block, test-locked.

The regimes, side by side. Left — one smile transported under different SSR values for the same spot move. Right — the ATM response $\Delta\sigma = \text{SSR}\cdot s_0\cdot\delta$ per regime: the number a vol trader quotes as "how much does vol come in if we rally 1%".

TRADING RELEVANCE

Scenario marks and priors move with spot **instantly and consistently with the desk's dynamics view** — recalibration then only absorbs genuine news.

Prior persistence: the fit remembers yesterday's smile shape, not its level — and only where today's quotes are silent.

The idea in one line: carry yesterday's surface into today's fit, but only where today's market says nothing. Mechanically, the prior adds a few soft pull-terms to the fit — "stay near yesterday's spot-transported value" — one per smile feature (ATM level, skew, curvature, wings). Each pull's weight is **gated**: exactly zero for any feature today's quotes already pin down.

ONE UNIVERSAL RESIDUAL SHAPE — MODEL VS TRANSPORTED PRIOR, PER FEATURE

$$r_j \;=\; \sqrt{\lambda_j}\,\frac{O_j(\text{model}) - O_j(\text{prior})}{\text{scale}_j}$$

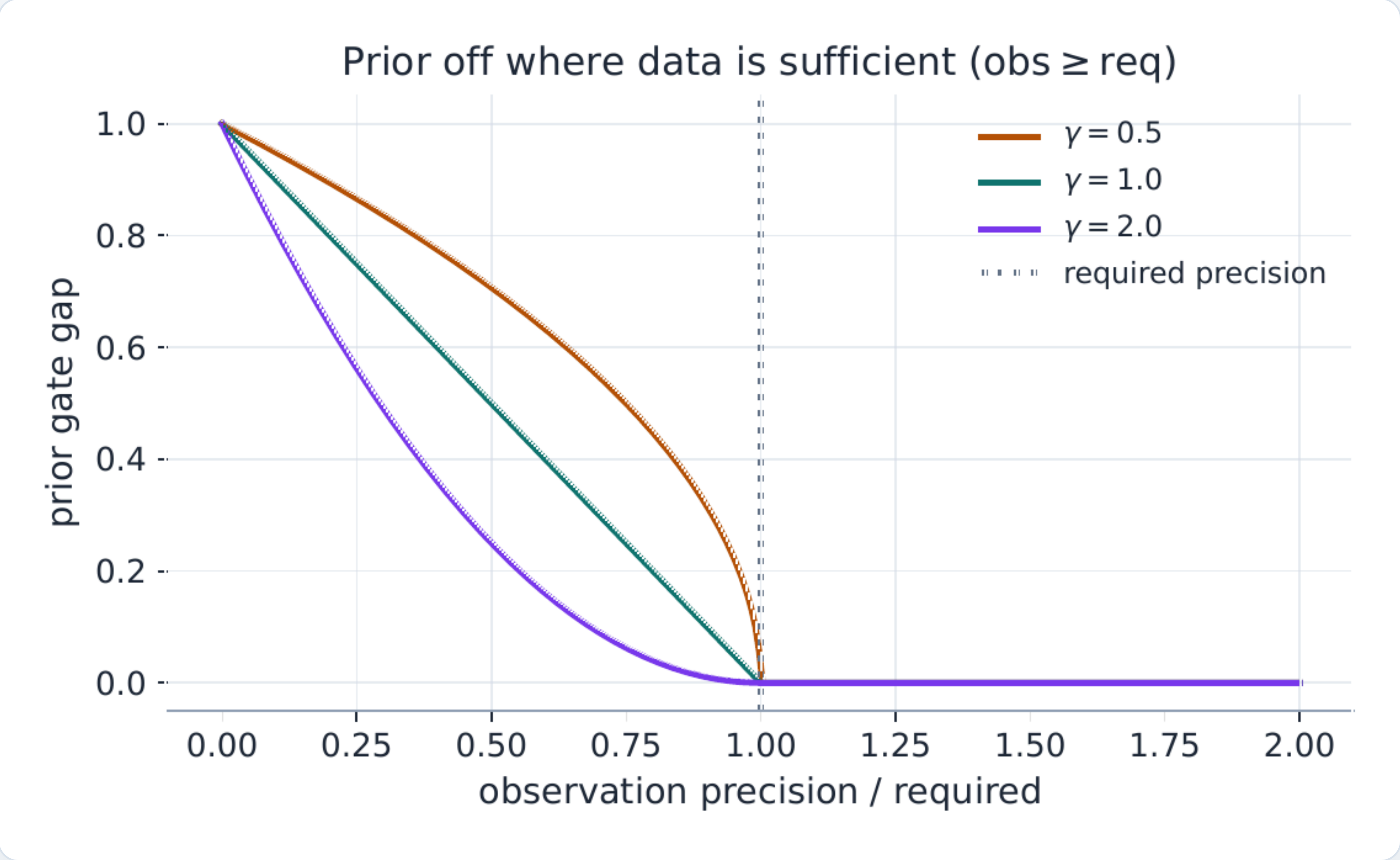
r_j — penalty row for feature j; O_j — feature functional (ATM/RR/BF/var-swap basket, smile factor, or delta-located price), applied to today's model and the transported prior; λ_j — gated weight; scale_j — natural scale.

THE ACTIVATION GATE — EXACTLY ZERO WHERE DATA IS SUFFICIENT

$$\text{gap}(\pi^{\text{obs}}) = \Big(\min\big(\max(1-\pi^{\text{obs}}/\pi^{\text{req}},\,0),\,1\big)\Big)^{\gamma}, \qquad \lambda_j = B\,\frac{\text{gap}_j}{\sum_i \text{gap}_i}$$

gap — prior activation in [0,1] (1 = data silent, exactly 0 once identified); π^{obs} / π^{req} — observed vs required feature precision; γ — sharpness of the on/off edge.

- **No-damp guarantee** — a feature today's quotes identify precisely is never pulled toward yesterday (proved, and test-locked).
- **Skew and curvature are remembered as baskets** — RR/BF combinations unchanged when the whole smile shifts level — so after an overnight jump the fit keeps yesterday's *shape* without fighting the new level.
- **Echo-proof:** graph output is never prior input — only lit, calibrated nodes enter a prior snapshot (certification-locked).



The gate, read left to right. x-axis — how precisely today's quotes pin the feature, as a fraction of the required precision; y-axis — how strongly the prior pulls (1 = full, 0 = off). With no quotes the prior is fully active; as quote support builds, the pull fades; at the required precision it hits **exactly zero and stays zero** — today's data always wins. γ sets how early the fade starts (γ = 2 lets go sooner; γ = 0.5 holds on longer).

TRADING RELEVANCE

Sparse afternoons stop destroying morning marks: **the surface holds its shape through quote droughts without ever fighting live information**.

Prior persistence, validated: seven modes raced on held-out wings — hybrid won and ships as default.

Which way of "carrying yesterday" actually helps? Seven candidate prior modes were raced on history: take consecutive-day pairs, fit day two with each mode while **hiding all of one wing's quotes**, and score each mode on how well its fit reconstructs the hidden wing. A mode wins only if yesterday's information genuinely fills today's gap. (RR/BF = risk-reversal and butterfly, the standard skew and curvature baskets.)

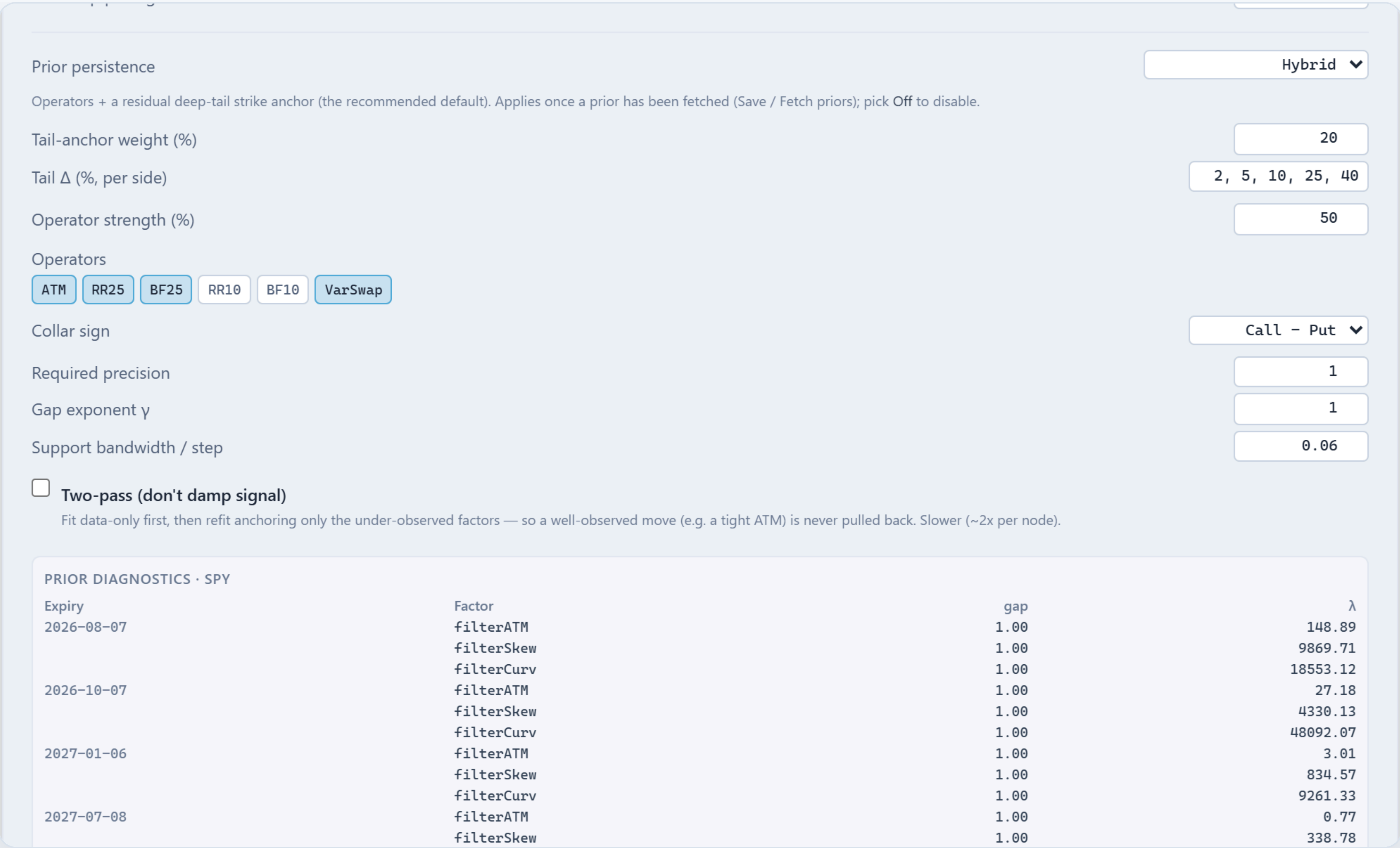
~32 bp

better held-out wing reconstruction vs no-prior (hybrid, full spike regime)

66%

win rate over 1,116 nodes — Aug-2024 spike regime, 8 assets, 19 day-pairs (cross-regime rerun queued)

- **The winner — "hybrid":** soft pulls on the smile features (ATM/RR/BF) plus one anchor deep in the tail. It beat every alternative at every prior strength tested — the shipped default is the measured winner, not a preference.
- Two of the seven modes — quote-operator and smile-factor pulls, level-invariant baskets that carry no absolute wing level — **never beat using no prior at all** at the median. That negative result is reported, not hidden.
- **Case file:** anchoring individual strike *levels* pulled a genuinely repriced surface halfway back to yesterday after an overnight jump. What ships instead: remember skew and curvature as level-invariant baskets — shape persists, level stays free.
- Every activation is **auditable**: a per-expiry table in the UI shows which prior acted, how strongly, and how much quote support the feature had.



The audit surface. The Calibration card: prior mode (Hybrid), tail-anchor and operator strength, operator selection (ATM/RR/BF/VarSwap), gate parameters, the two-pass "don't damp signal" option — and the live per-expiry prior diagnostics table.

TRADING RELEVANCE

Memory with a paper trail: **which prior acted, where, and how strongly is a table in the UI** — not a mystery inside the optimizer.

DYNAMICS THE OBSERVATION FILTER — NOISE VS NEWS

The observation filter — a Kalman filter in handle space — separates quote noise from genuine surface motion.

Two consecutive calibrations never agree — some is the market moving, most is quote noise. Each calibration is a noisy **measurement** of the true handles (ATM vol, skew, curvature); yesterday's filtered state, moved by the SSR transport, is the **prediction**; relative precision blends them via the **Kalman gains K ∈ [0,1]** — K = 1 "believe today fully", K = 0 "today is noise".

THE UPDATE — RELATIVE PRECISION DECIDES HOW MUCH OF TODAY TO BELIEVE

$$S_t = H P_t^- H^\top + R_t, \qquad K_t = P_t^- H^\top S_t^{-1}, \qquad m_t^+ = m_t^- + K_t\,(z_t - H\,m_t^-)$$

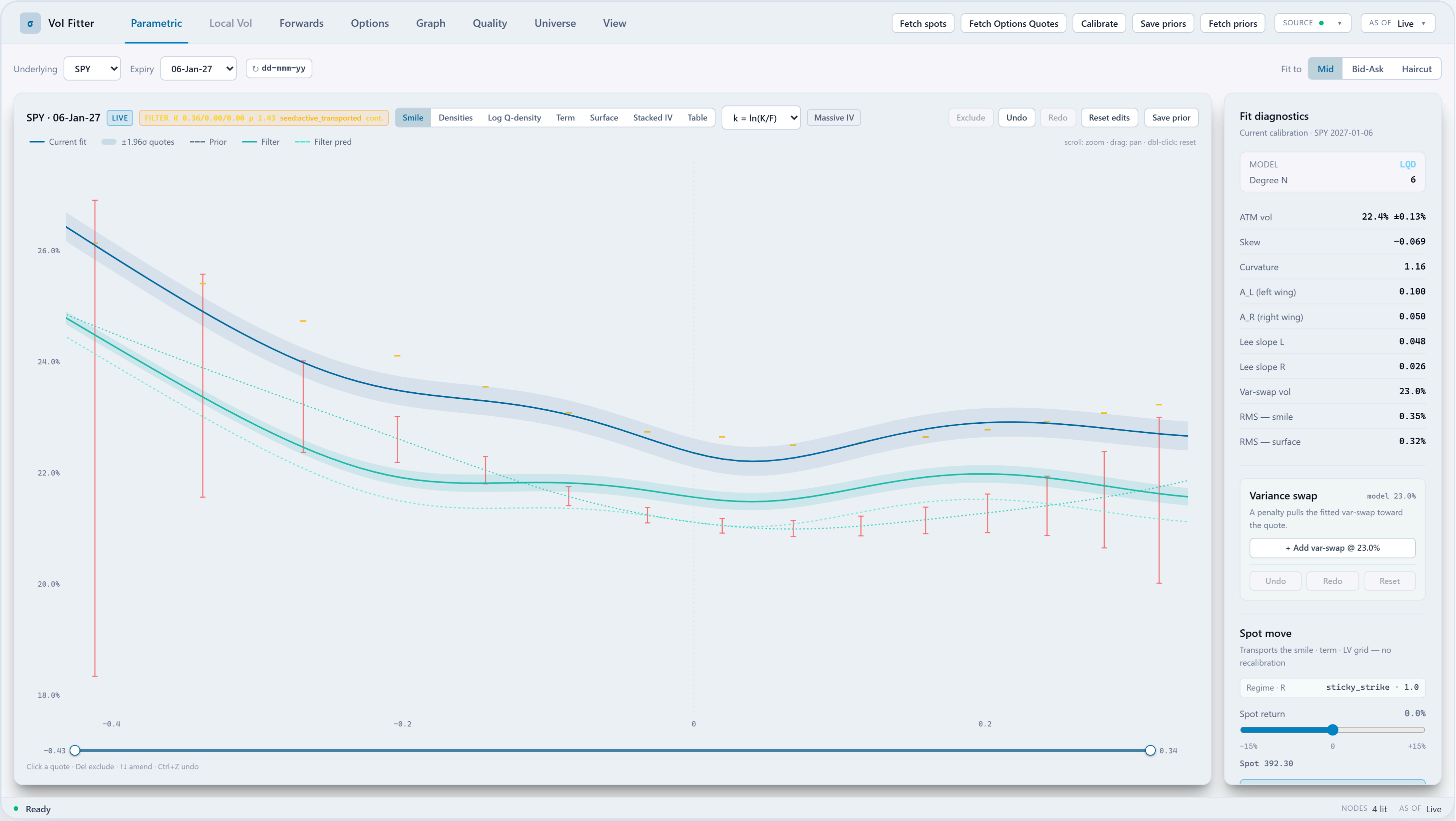
m^-, P^- — predicted (transported) handle mean/covariance; z — noisy handle observation; H — observation operator; R — measurement covariance; S — innovation covariance; K — Kalman gain; m^+, P^+ — filtered mean/covariance.

ACTIVE MODE — A ONE-STAGE MAP FIT (PROVABLY NO DOUBLE-COUNTING)

$$\mathcal{L}_{\text{active}} = \mathcal{L}_{\text{quote}} + \tfrac{1}{2}\left\|\mathcal{H}(\theta) - m_t^-\right\|^2_{(P_t^-)^{-1}} + \mathcal{L}_{\text{intrinsic}}$$

θ — model parameters; L_{quote} — misfit to today's quotes; H(θ) — handle read-off; m^-, P^- — prediction mean/covariance (penalty weight $(P^-)^{-1}$); $L_{\text{intrinsic}}$ — model regularizers; $L_{\text{persist},\perp}$ — persistence on coordinates the filter does not carry.

- **Both noise levels are measured, not assumed:** today's noise is the fit's own reported uncertainty, widened when its residuals run worse than usual; the drift allowance was set by backtest, not taste.
- **The drift allowance is adaptive:** on genuine shocks it opens so the filter follows the market (prediction errors of 39–79 bp fall to 5–8 bp) — and stays closed on quiet days.
- **Two clocks for the prediction law:** calendar (default) or an opt-in session clock — an overnight and a whole *weekend* both move ATM ≈ 55 bp, 30 session minutes 19.5 bp; only a session clock fits all three.



today's raw fit — filter posterior (shaded: ±1σ) — prediction (yesterday, transported) — bid/ask quotes

The filter on a smile (controlled demo session; amber ticks mark quote mids that were manually amended to stage the move). The badge is the audit: per-handle gains K = 0.36 / 0.00 / 0.00 and residual inflation ρ = 1.43 — the filter believes

TRADING RELEVANCE

Marks stop chasing microstructure: **the surface moves when the market moves, and holds when only the quotes wiggle.**

The observation filter was tuned by 39,190 backtested steps — its best mode beats both of the inputs it blends.

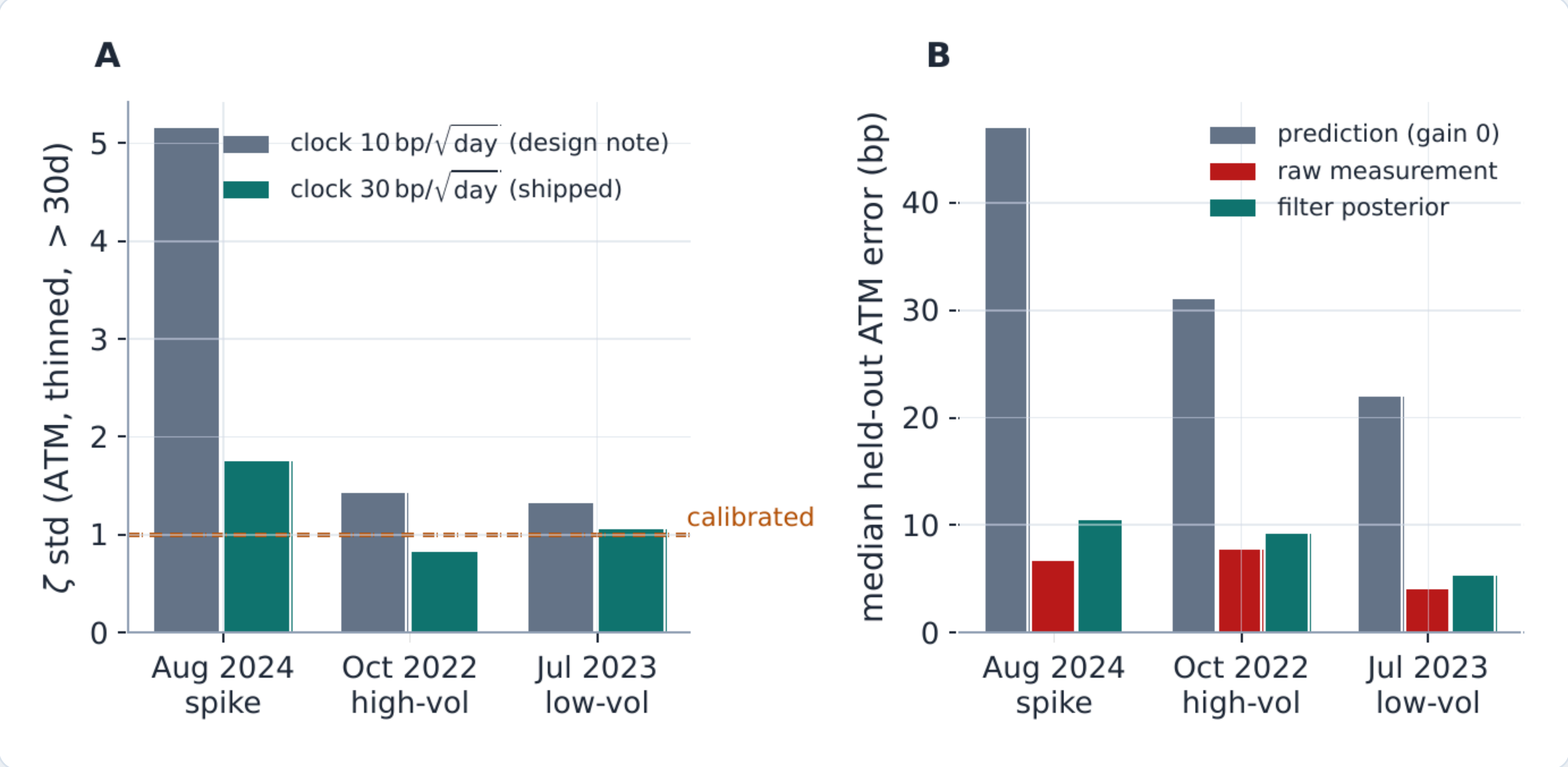
None of the filter's settings were chosen by taste: three historical regimes — the Aug-2024 spike, high-vol Oct-2022, low-vol Jul-2023 — were replayed day by day, 39,190 filter steps, and each design question was settled by the errors. The strongest mode is **"active MAP"** (maximum a posteriori — the most probable state given both sources): rather than fitting today's quotes first and averaging with the prediction afterwards, the prediction enters the fit itself as extra penalty rows, so quotes and prediction are weighed **once, inside one optimization** — provably equivalent to the Kalman update, with no quote double-counted.

5.6 vs 9.8 bp
active MAP vs raw fit on contradiction days (high-vol regime)

39,190
filter steps backtested — 3 regimes × 8 assets × all day-pairs

10 → 30 bp/√day
process-noise default flipped by the evidence (one-sided win)

- **Active MAP is the best denoiser on thinned days in every regime** — more accurate than the raw fit it takes in *and* than blending after the fit.
- **Its measured weakness is outright shock days**, where active still trails the raw fit (19.5 vs 3.8 bp, spike; 25.2 vs 5.8, high-vol): the adaptive drift-gate is overlay-only today — closing that is the stated blocker before active-by-default.
- **Case file:** on coarse ETF chains (EEM/EFA), letting all handles update *together* let junk curvature drag the ATM level — errors up to 28 vol points. Production updates each handle independently.
- **Case file:** two stale quotes contradicting the chain read gains 0.80/0.72/0.03 — absorbed as noise; when the whole chain repriced, the gate opened and the filter followed. Noise resisted, news respected.



Two panels, two questions. A — are the uncertainty bands honest? ζ = error ÷ its own predicted uncertainty; honest bands mean spread ≈ 1. The old 10 bp/√day drift budget reads 5.2 in the spike (bands far too narrow); the shipped 30 bp/√day sits near 1 (1.8 in the spike) — which flipped the default. **B — does filtering cost accuracy?** Yesterday's prediction alone misses by 22–47 bp; today's raw fit and the filter posterior both land under ~10 bp — stability and honest bands at ~no accuracy cost, and on contradiction days the filter wins outright (5.6 vs 9.8 bp).

TRADING RELEVANCE

The filter's gains are a **per-expiry trust report on the live market** — a low K on curvature is the system telling you that part of the smile is noise today.

GRAPH THE PROPAGATION LAYER — CONCEPT

Graph extrapolation: the universe is a graph of smiles — with three propagation modes, ordered by how much they assert.

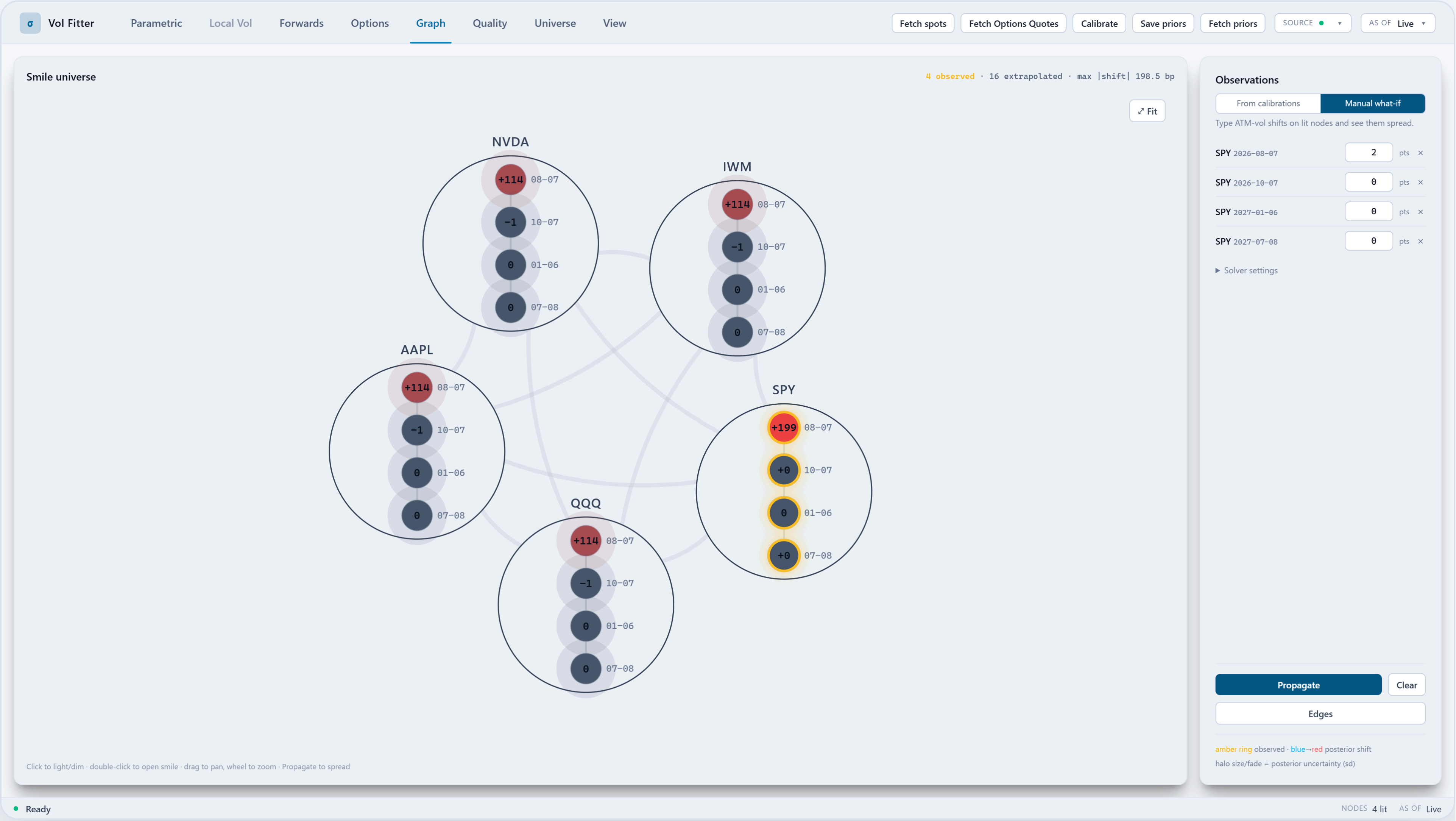
Nodes are (underlying, expiry) smiles. **Lit** nodes have usable quotes and get calibrated; **dark** nodes do not — they are the point of the exercise. Edges say which smiles should co-move: adjacent expiries of one name (calendar edges) and same-tenor smiles of related names (cross-asset edges). What flows over them is a choice of **assertion**: smoothness, contracts, or dependencies.

THE SMOOTH-FIELD MODE: ROW-STOCHASTIC KERNEL, CONDUCTANCES, B-WEIGHTED DIRECTED LAPLACIAN

$$K_{ij} = \frac{w_{ij}}{\sum_\ell w_{i\ell}}, \qquad c_{ij} = \tfrac{1}{2}\left(\pi_i K_{ij} + \pi_j K_{ji}\right), \qquad L^{\beta}_{\mathrm{dir}} = (I - K^{\beta})^{\top}\Pi\,(I - K^{\beta})$$

$w_{ij} \geq 0$ — raw weight "node j informs node i"; K — row-stochastic trust kernel; π — stationary distribution of K (Π = diag π); c_{ij} — reversibilized conductance; β — per-edge amplitude matrix (K∘β — entrywise product); L_{dir}^{β} — β-weighted directed Laplacian; I — identity.

- Nodes are **(underlying, expiry)** smiles; the lattice is what the desk already draws on a whiteboard — SPY 1M talks to SPY 3M and to QQQ 1M.
- **Three modes, one solver family:** *smooth field* (assert smoothness — the Laplacian above), *precision messages* (assert relations — **the production default since 2026-07-27**), and *layered dynamic-harmonic* (assert dependencies — research-grade, deliberately opt-in). Mode changes are explicit and evidence-gated.
- All three regularize **increments, not levels** — "no change from the transported prior" is always the default; the graph never invents a move. Lit vs dark is a per-node switch on one universe object.



The lattice (controlled demo session). Tickers × expiries, lit (calibrated) nodes ringed amber. A unified what-if lets a trader light nodes, pulse hypothetical innovations and watch the posterior field respond before committing anything; production runs replace the pulses with real calibration innovations.

TRADING RELEVANCE

The graph formalizes what traders do informally — **"SPY repriced, so my dark midcap should too, a little"** — with a posterior and a band instead of a hunch.

The default operator: every edge is a contract — precision messages, with exact attribution.

Innovation = what today's calibration says minus what yesterday's spot-transported surface predicted — the genuine news in a lit name. Only that increment propagates; levels never do. In the default **precision-message** mode, each edge is one Gaussian pairwise factor: an **amplitude β** (the contract — how many units of the neighbour's move a node inherits) and a **precision p** (the trust — measured, never asserted). Amplitude and trust never mix.

ONE PAIRWISE RELATION FACTOR PER EDGE — PSD FOR ANY AMPLITUDE, SPARSE BY CONSTRUCTION

$$Q_{\text{msg}} \;=\; \sum_{(i,j)\in E} p_{ij}\,(e_i-\beta_{ij}e_j)(e_i-\beta_{ij}e_j)^{\top}, \qquad \beta_{ij}^{\text{cal}} = \left(\frac{T_j}{T_i}\right)^{\alpha_T}, \quad \alpha_T = 1$$

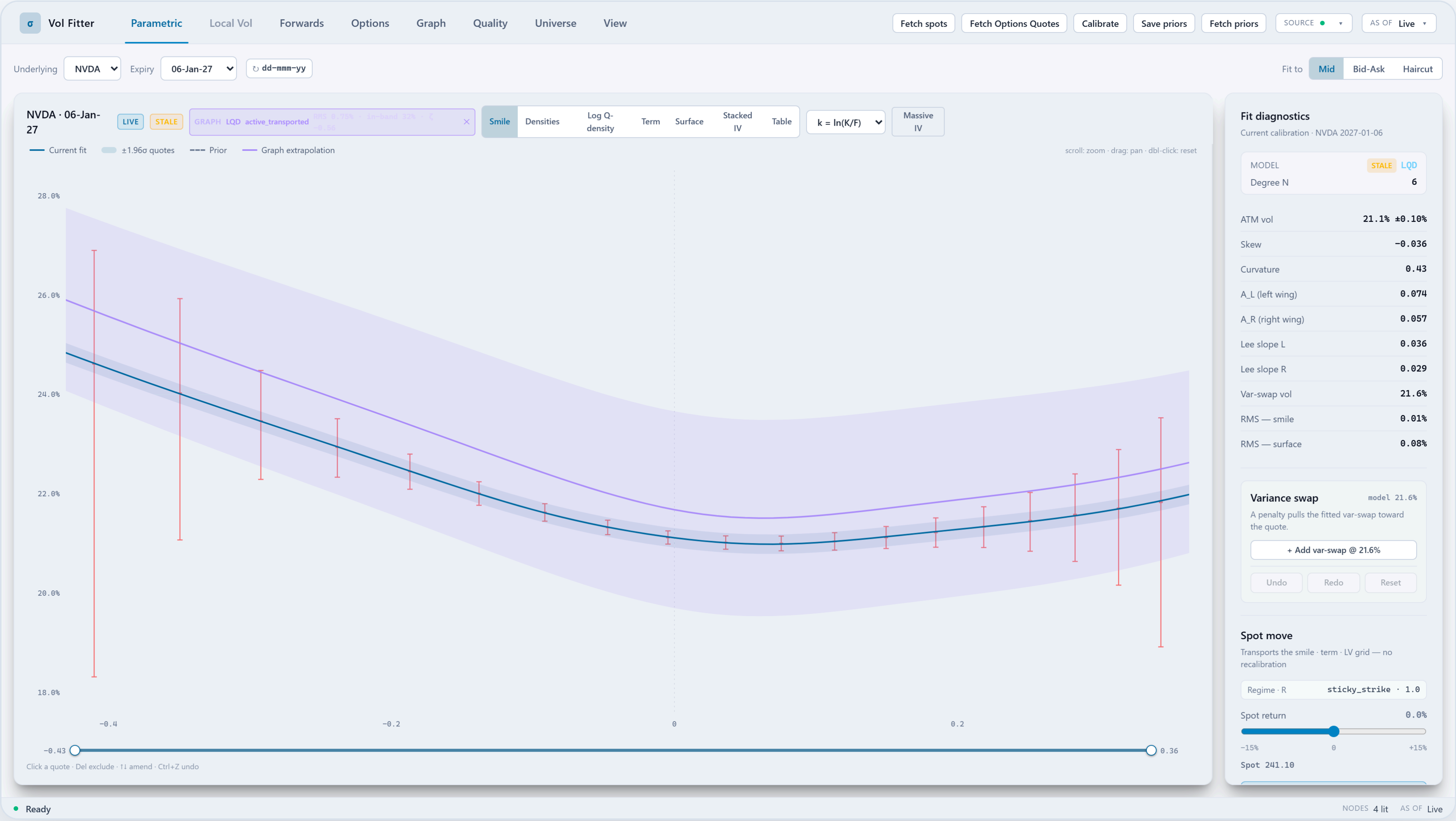
Q_{msg} — message precision (Gaussian prior on the innovation field); E — edge set; p_{ij} — edge precision (trust, measured from data); β_{ij} — edge amplitude (the contract); e_i — coordinate selector of node i; calendar amplitude $(T_j/T_i)^{\alpha_T}$ with $\alpha_T = 1$ — constant total-variance transport along the ladder.

INFORMATION-FORM POSTERIOR — AND PER-SOURCE GAIN ROWS THAT SUM TO THE MOVE EXACTLY

$$Q^{+} \;=\; Q_{\text{msg}} + D_{\kappa} + H^{\top}R_d H, \qquad \hat{z} \;=\; (Q^{+})^{-1}H^{\top}R_d\, d \;=\; \textstyle\sum_{s\,\in\,\text{lit}} g_s\, d_s$$

Q^{+} — posterior precision; D_{κ} — per-node anchor to the transported prior ("no news" ⇒ no move); H, R_d — lit-node selector and observation precisions; d — observed lit innovations; $\hat{z}$ — posterior innovation field; g_s — per-source gain rows: every dark move decomposes into named lit-node contributions, summing exactly by construction.

- **Factors, not averages:** a dark node in the middle of a chain cannot dilute a message — the "dead informer" case is a golden test, not a hope.
- **Containment:** innovations enter weighted by the lit fit's own precision; components with no lit path stay at *zero* innovation with deliberately wide bands — no artificial precision is invented. Every observed component is



graph posterior (shaded: credible band) node's own last calibration bid/ask quotes (not used — node is dark)

A dark node, reconstructed (controlled demo: only SPY is lit, its quote mids amended +150 bp). NVDA's six-month smile is unquoted in this session. The violet posterior lifts it **+57 bp** off its transported prior by the graph-propagated innovation, inside

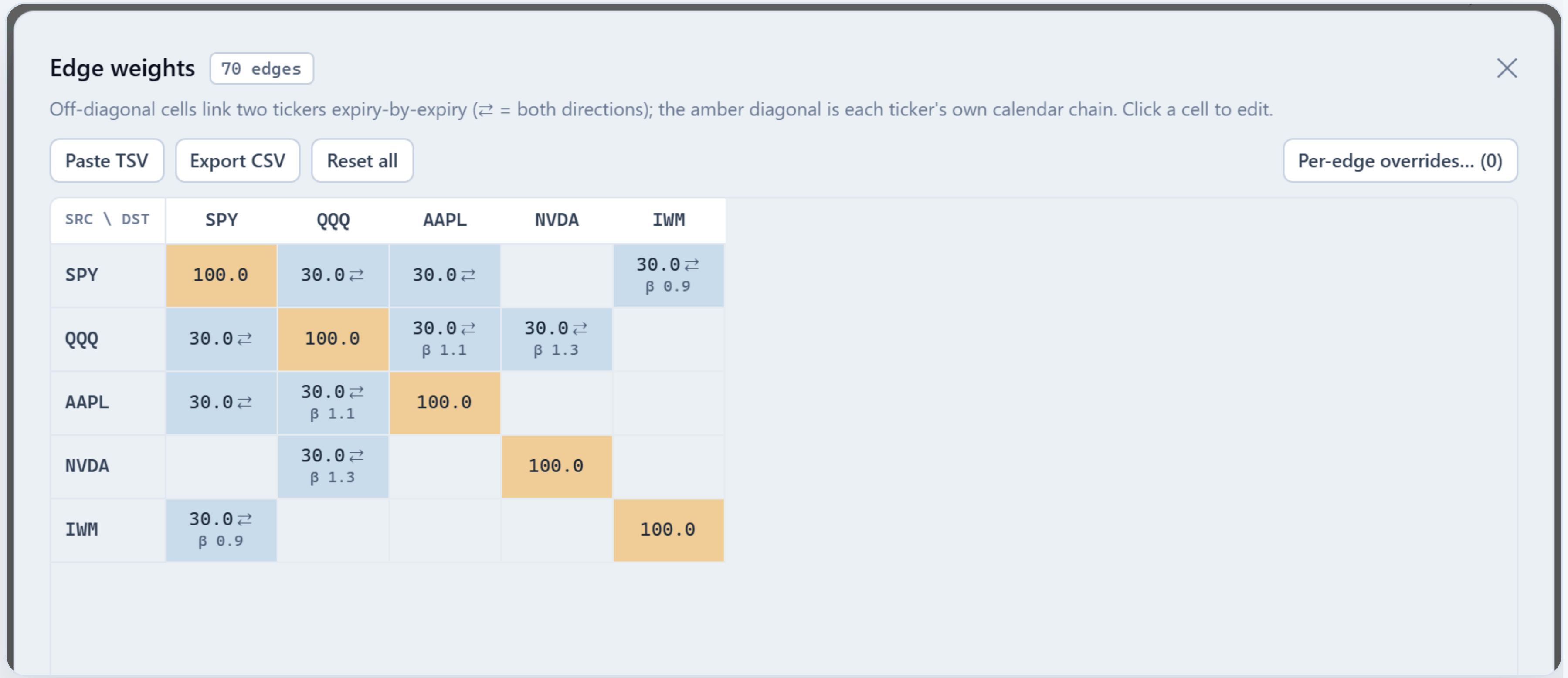
TRADING RELEVANCE

Sparse single names inherit **information, with uncertainty attached** — not levels — from liquid neighbors.

Every graph edge is a number a trader can defend — weights, betas, and an editor to own them.

Propagation strength is not a black box: each edge carries a weight (how strongly two smiles are coupled) and a **β** (how many units of the neighbour's move a node inherits — the same β a trader means by "NVDA moves 1.3× QQQ").

- **Per-edge β** scales how much of a neighbour's innovation a node inherits — index→name edges can carry β > 1 (levered names) or β < 1 (defensive). The calendar *shape* is locked ($(T_j/T_i)^{\alpha T}$); the *level* is desk-set at 1.0, with historically-learned presets selectable.
- **Edge policies are first-class objects** — a policy card with **live preflight** (blockers named before any solve), a draft→active lifecycle, a per-edge message inspector, and a validation drawer that runs the production LOO side-by-side before adoption.
- **The reach autotune is the production LOO harness itself** — the tuned dial optimizes exactly the objective the backtest reports (the old synthetic sandbox tuner was deleted outright).
- Every dark-node move **decomposes exactly**: the attribution card lists per-neighbour contributions (gain × innovation) that sum to the posterior shift — test-locked, not approximate.
- Confidence honesty is a stated caution: bands come from the **marginal posterior covariance**, never the precision diagonal.



The edge editor (controlled demo session). The persisted per-edge view over the selected universe: seeded defaults, editable weights and β, cross-ticker toggles — opened directly from the Extrapolate panel that consumes it.

β per edge

directed, PSD-safe for any amplitude (proved), persisted with the universe

LOO reach

cross-validated on the lit nodes, one click, production objective

TRADING RELEVANCE

The propagation model is **auditable in the desk's own language** — "NVDA follows QQQ at β 1.3" is an editable row, not a latent parameter.

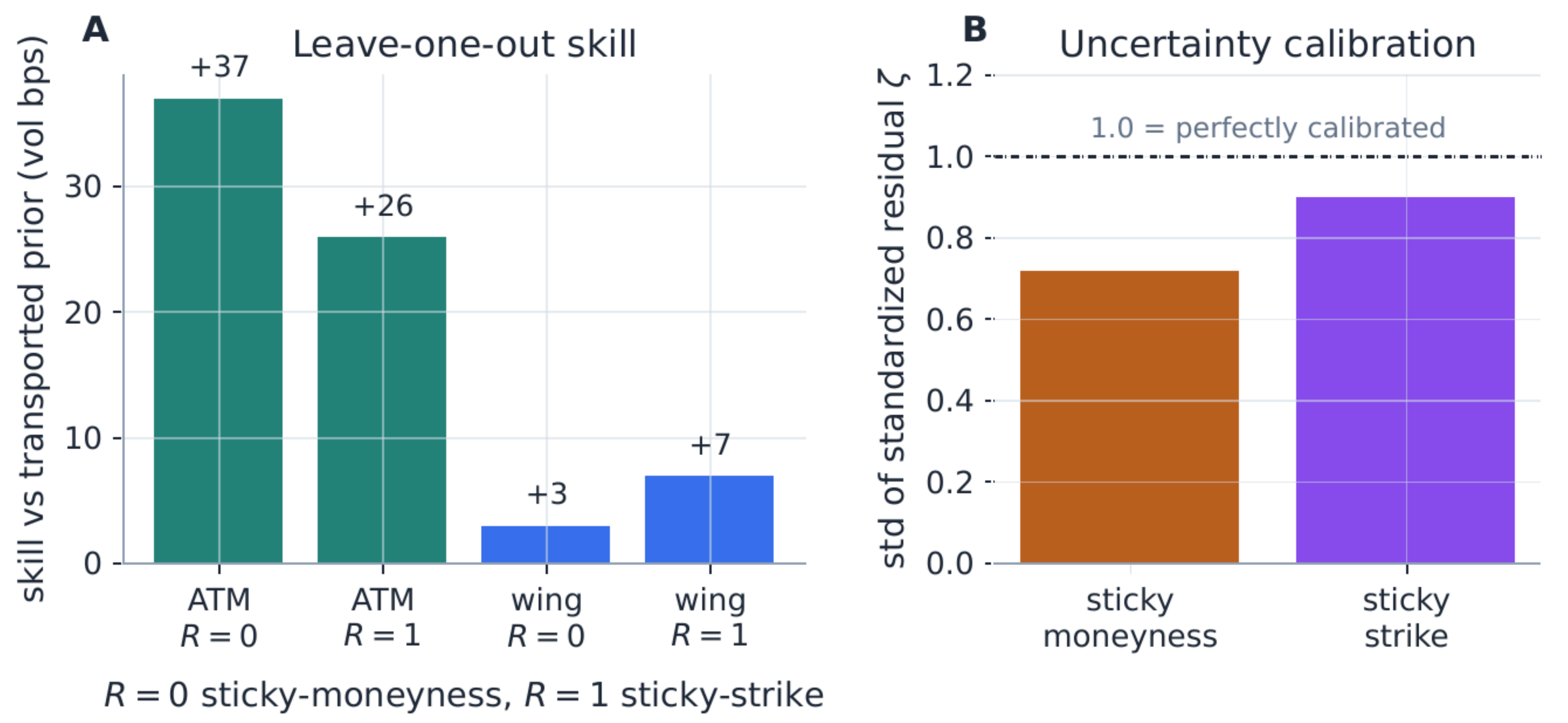
Graph skill is demonstrated on held-out history — including fully-dark names in the regimes that matter.

Leave-one-out (LOO): pretend a lit node is dark, let the graph reconstruct it from the others, then compare to the calibration it actually had. The baseline to beat is the mechanical alternative — just spot-transporting yesterday's surface. "Skill" = how much closer the graph gets. 25 assets × 3 historical regimes, 47,393 held-out scores.

+8…+14 bp
fully-dark single names behind lit indexes/ETFs — Aug-2024 spike, the product case

+3.8…+7.2 bp
same, fully out-of-sample (Oct-2022 bear; reach tuned on the spike only)

- **Neighbour-supported holdouts: decisive skill, calendar-driven** — indexes +10…+76 bp, ETFs +3…+7 bp across all three regimes (ranges bracket the SSR transport assumptions).
- **Honest scope:** the dark-name numbers are at assertive reach (η 10, cross-conductance ×25 — tuned on the spike; the shipped default reach is deliberately timid, ≈+0.1 bp). Reach is a desk dial, and the one-click LOO is how a desk picks it with evidence. In the Jul-2023 calm the skill is ≈0 (never negative) — earnings-idiosyncratic tape carries no systematic signal.
- **The bands are honest in every regime:** after an idiosyncratic band floor (a name's band can never fall below ~0.55× its own trailing innovation RMS), ζ std reads ~1.0 in all three regimes — the calm cell's 1.9 overconfidence was found by this audit and fixed by it.



Leave-one-out, plotted (8-asset pilot run; the 25-asset benchmark ships as a regenerable HTML artifact). Held-out node predictions versus realized calibrations: the graph posterior (vs the transported-prior baseline) tightens errors and its standardized residuals sit where a correct model's should.

TRADING RELEVANCE

A propagation engine you can **re-certify on your own book any day** — it earns its keep exactly when the market reprices together, and its weakest case (dark names in calm, idiosyncratic tape) is reported at its measured ≈0, not claimed.

Defaults flip on pre-registered evidence — the message operator earned its seat in an intraday replay.

Every propagation default faces a gate **written before the sweep runs**, and holds are recorded as loudly as wins. The deciding campaign: an **asynchronous intraday replay** over 8 captured sessions of an SPY/QQQ/IWM triangle (312 NBBO snapshots, 13 instants a day) — one ticker is relit mid-session and its dark neighbours are scored before and after, 13,356 rows per arm.

PROPAGATION ARM	ATM RMS BP	Z STD	95% COV
pure transport (no graph)	172.7	–	–
smooth field	168.6	0.57	1.00
precision messages	65.8	0.88	0.96
layered · best memory (½-life ≈ 2.4 h)	73.7	0.80	0.97
layered · undecayed memory (H = ∞)	108.1	2.75	0.54

Dark-neighbour ATM error after the mid-session repricing. The smooth field is nearly inert intraday (its day-scale anchors shrink intraday innovations to nothing); the message operator carries the news at 2.6× lower error with honest bands — the separation the default flip was ratified on.

- **The flip, scoped precisely:** precision messages became the workstation default on 2026-07-27, user-ratified on this separation; the smooth field remains the explicit rollback and the untouched replay/harness default — byte-identity locks intact.
- **Its own daily-horizon gate table is filled and published:** at one-day granularity the gain is real but immaterial (+0.4…+2.0 bp) and the graph credible bands read overconfident (ζ ~2) — recorded verbatim; band calibration is the named next fix, not a footnote.
- **The layered mode was validated and still held:** its residual memory shows a genuine interior optimum (½-life ≈ 2.4 h, persistence skill decaying 18 → 2.5 bp, calibrated ζ 0.80) — but the static message operator beat every layered arm, so it stays opt-in. The verdict, the reason, and the decisive follow-up experiment are all on file.

TRADING RELEVANCE

Model governance you can diff: **gates written before the sweep, RECORD/HOLD verdicts, explicit rollbacks** — the propagation layer is adjudicated with the same discipline as the fits.

PRODUCT UNCERTAINTY AS AN OUTPUT

Every mark ships with its error bar — and the posterior answers "what should we quote next?"

Uncertainty here is not a shaded region drawn after the fact — it is computed from the same objects that price, audited in backtests, and turned into product features of its own.

Functional credible bands

The full 3-handle posterior covariance is pushed through the slice map (delta-method): skew and curvature uncertainty **widen the wings**, level uncertainty widens the money — and the same six perturbed slices price **var-swap and tail-mass standard deviations** carried on every payload.

Error bars from the quotes

Every calibration keeps its solver Jacobian; with the **bid–ask half-spread as the stated per-quote noise**, each fit prices its own parameter and handle uncertainty — even with the filter off. Advisory by construction: it never touches a calibration.

Observation planning · closed form

A rank-one update of the graph posterior ranks every candidate dark node by **expected variance reduction** — "which node is most worth a quote" — with **no refit and no re-solve**, optionally exposure-weighted. One endpoint, computed from what the solver already knows.

ζ, the audit

Every band the system quotes is scored in backtests by standardized residuals. Band honesty (ζ std ≈ 1) is a **tracked, certified metric** — and dishonest cells get fixed (the calm-regime idio floor), not excused.

±1.96σ

credible band on every dark reconstruction — judged against realized calibrations later

0

refits or re-solves to rank every "quote next" candidate — rank-one algebra on the posterior

ζ ≈ 1

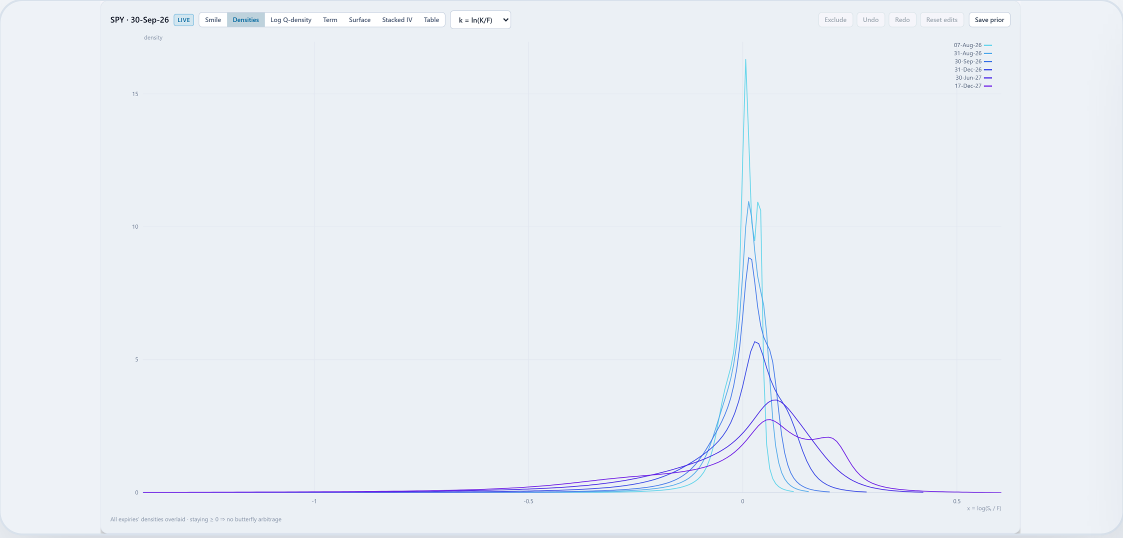
band honesty held in every backtested regime, locked by the certification pack

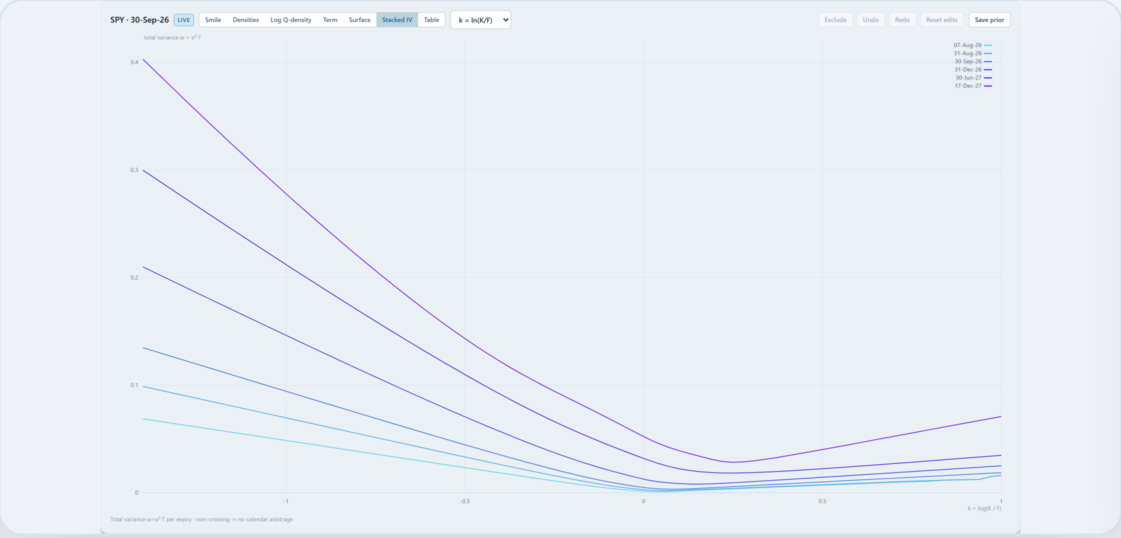
TRADING RELEVANCE

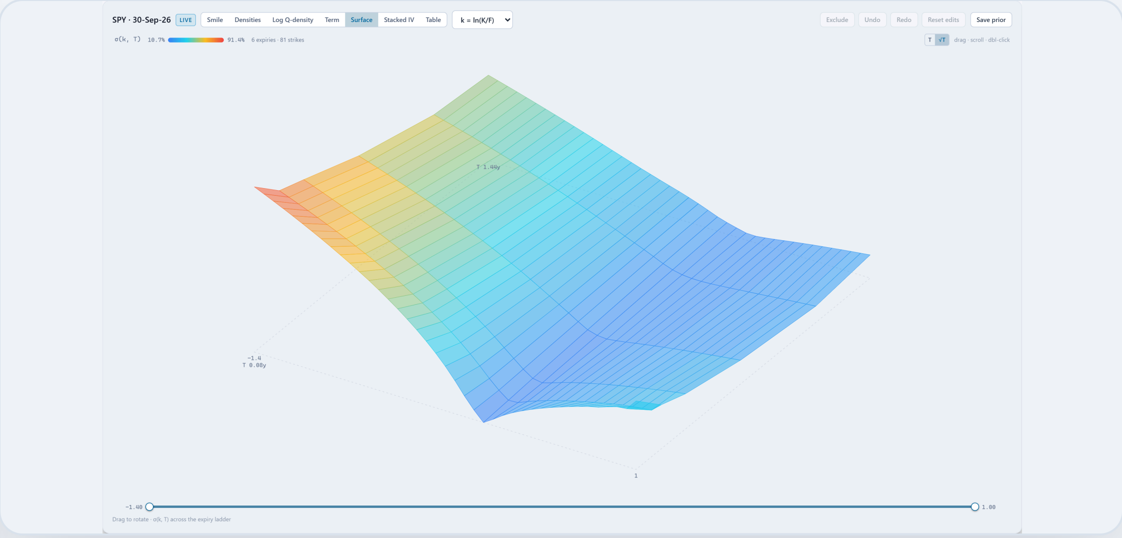
Uncertainty is not decoration: **bands gate publishing, weight the graph's trust, and rank the desk's next quote**.

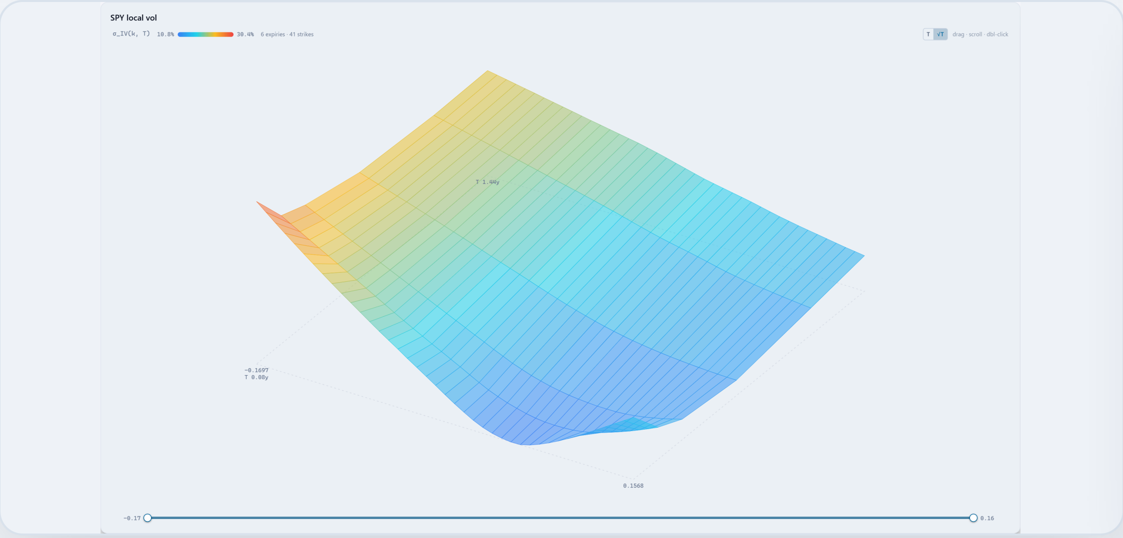
PRODUCT THE WORKSTATION TODAY

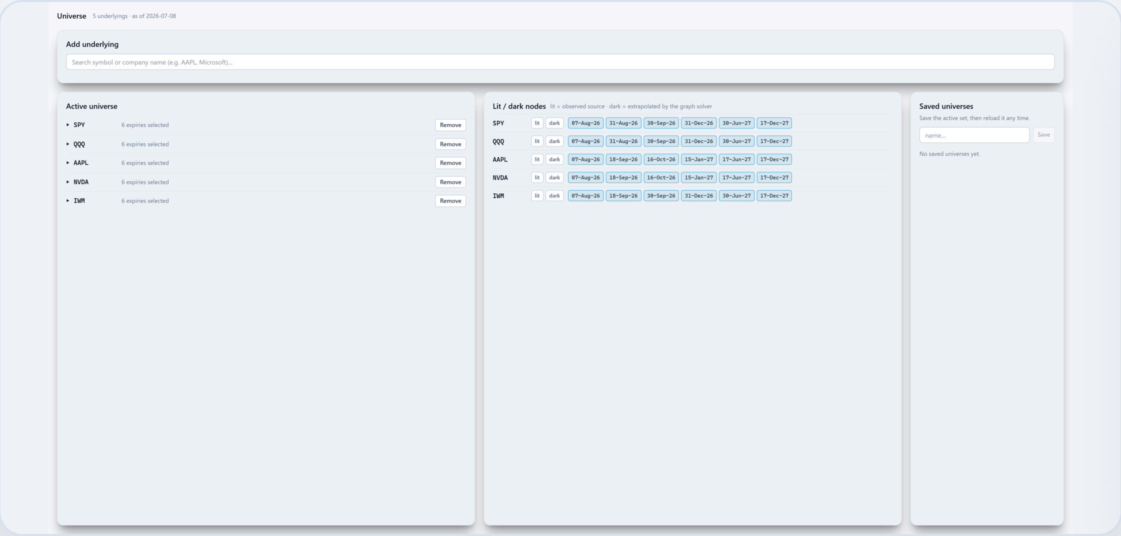
Everything above ships in a working workstation — eight workspaces, live against three data feeds.

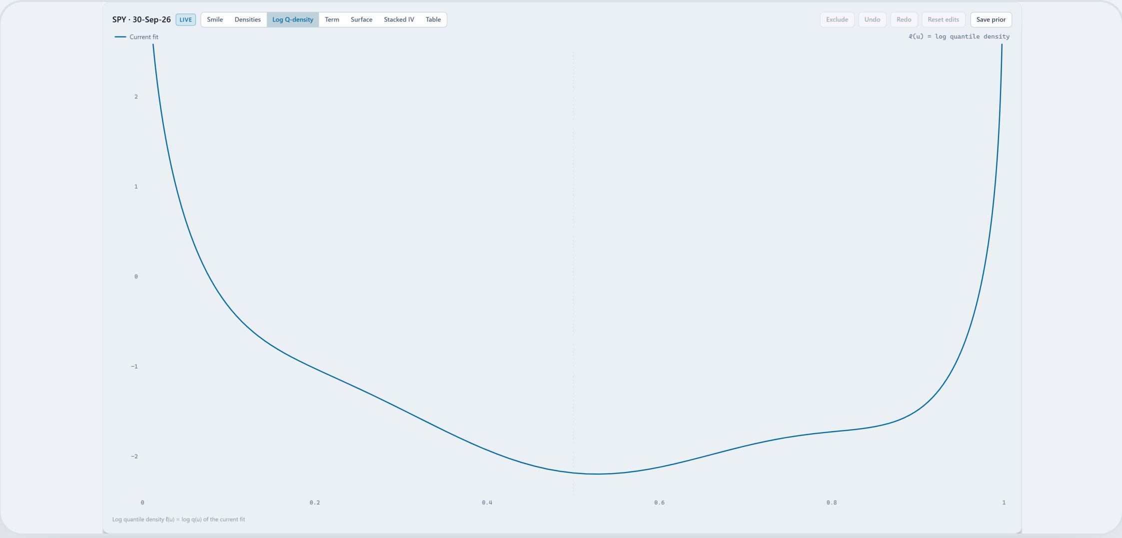

Stacked densities — every expiry ≥ 0: no butterfly arb, at a glance


Stacked total variance — non-crossing: no calendar arb


3D IV surface — rotatable σ(k,T) mesh


Local Vol views — surface, smiles, tables off one calibrated object


Universe manager — catalogue, expiries, lit/dark matrix, saved universes


Log quantile density — the model's native coordinate, charted

TRADING RELEVANCE

Parametric · Local Vol · Forwards · Options · Graph · Quality · Universe · View — **every modeling claim in this deck is a screen a trader can open**, live against Yahoo, Bloomberg or Massive.

PRODUCT FIT QUALITY AS A FIRST-CLASS SCREEN

The daily loop ends at a publish gate — fit quality is a dashboard, not a feeling.

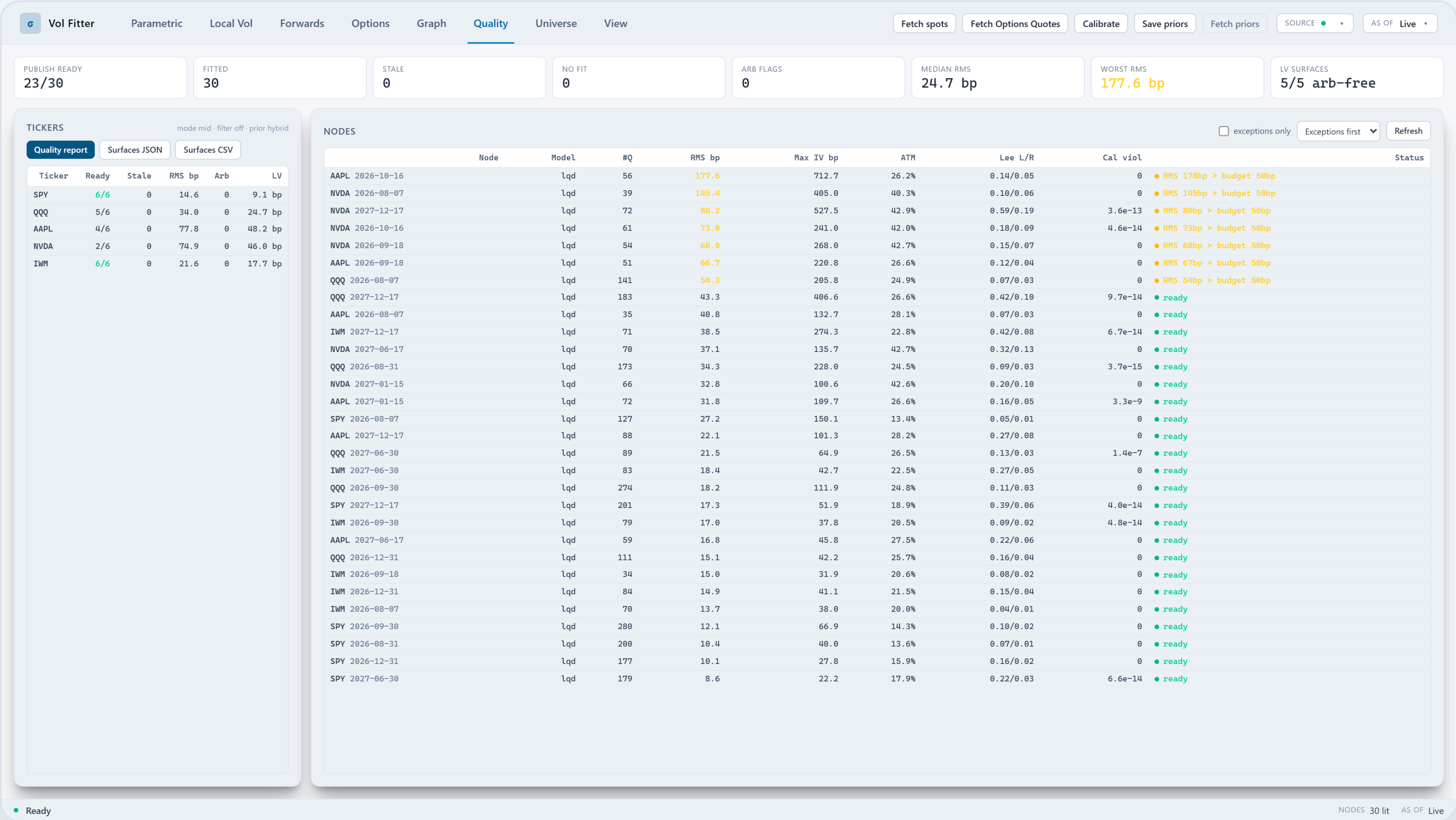
1 · Select & fetch
universe, expiries, lit/dark designation; spots + chains from the active source

2 · Calibrate
background per-ticker groups on a process pool (~2× warm) — byte-identical to serial

3 · Inspect & adjust
smiles, densities, term, LV; amend quotes, var-swaps, events, priors, filter

4 · Propagate
graph extrapolation over the dark universe, LOO-checked

5 · Gate & publish
per-node RMS budgets, butterfly certificates, data-age staleness — an uncertified belly or red-stale chain hard-blocks publish (HTTP 409, before any lineage is written); then JSON/CSV export + HTML report



The Quality workspace — live session (Massive feed). Publish-ready counts, per-node RMS against its budget, max-IV deviations, Lee slopes, calendar-violation flags and local-vol arbitrage status — exceptions first, one refresh, with the offending node one click away.

TRADING RELEVANCE

The question "can we publish this surface?" gets **a per-node answer with named exceptions — and hard blocks with reasons, not warnings**; degraded chains serve the transported prior *labelled with why*, never a silent blank. The runbook a vol desk actually operates.

ENGINEERING MEASURED, THEN OPTIMIZED

Speed is a feature with a budget: slice < 50 ms, surface < 1 s — met, and held by perf rails in the test suite.

Context for every number on this slide: single node = one (name, expiry) smile fit; measured on a laptop-class i7-12700H / 16 GB (the development machine — no server hardware involved). Live measurement, this deck's 30-node / 5-ticker Yahoo session: **full recalibration 7.6 s warm; 96.6 s cold start** (process-pool spawn, de-Americanization, kernel compilation — each cached thereafter).

96 → 35 ms	**~3 ms**	**~10 ms**	**~20 ms**	**~0.7 s**
LQD slice fit (2.7×) — "the atom of every parametric fit"	de-Americanization of a ~300-quote chain (Numba CRR; ~350 ms NumPy)	structural SVI fit — 3× vs raw, analytic Jacobian 2.1–2.4× again	warm 0DTE slice against a 50 ms design rail	graph posterior at 1,000 nodes (< 1 s target)

- **Measure first** is a design rule: every headline number above began as a profile, shipped as a fix, and is locked by a perf rail in the test suite (9 perf tests, budgets stated with hardware and cache state).
- **Analytic everywhere it pays** — LQD's implicit-root Jacobian, SVI Jacobians on both charts, MCS (3.4×), the FD-free arbitrage metric, closed-form var-swaps — and a vectorized safeguarded-Newton vol inverter worth ~39× on a curve render.
- **Compiled kernels with graceful degradation** — the LV march (~6× vs LAPACK banded, identical to ~1e-15) and the CRR tree run under Numba, each with a NumPy fallback; missing numba never breaks a fit.
- **Caches are content-addressed** — de-Am results keyed by chain digest; fit-tuning never re-pays preparation; invalidation is per-ticker, never global.
- **Failed optimizations are recorded too** — thread parallelism (a net loss under Python's global interpreter lock), coarse-grid warm starts (they broke the affine surface), and the Rannacher time scheme (~1.1× net, and non-monotone) were measured, rejected, and documented so they are not retried.
- Per-ticker process-pool calibration (~2× warm) is **byte-identical to serial** — speed never changes a number.

TRADING RELEVANCE

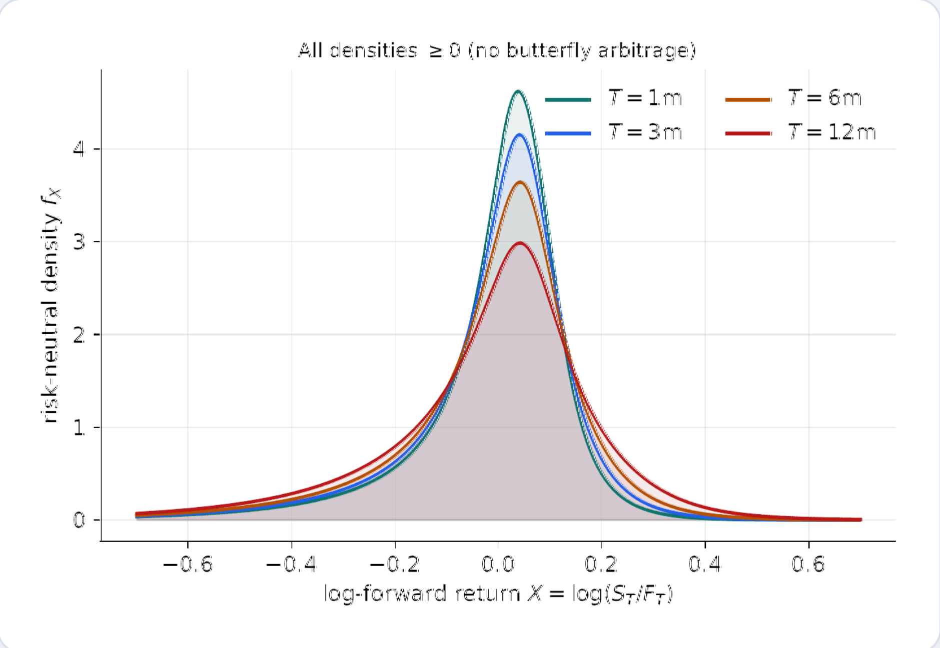
Fast enough to sit **inside the quote loop**: a 30-node universe re-marks in ~8 seconds warm on a laptop, a single slice in the time a quote blinks — and every number above states its hardware and cache state.

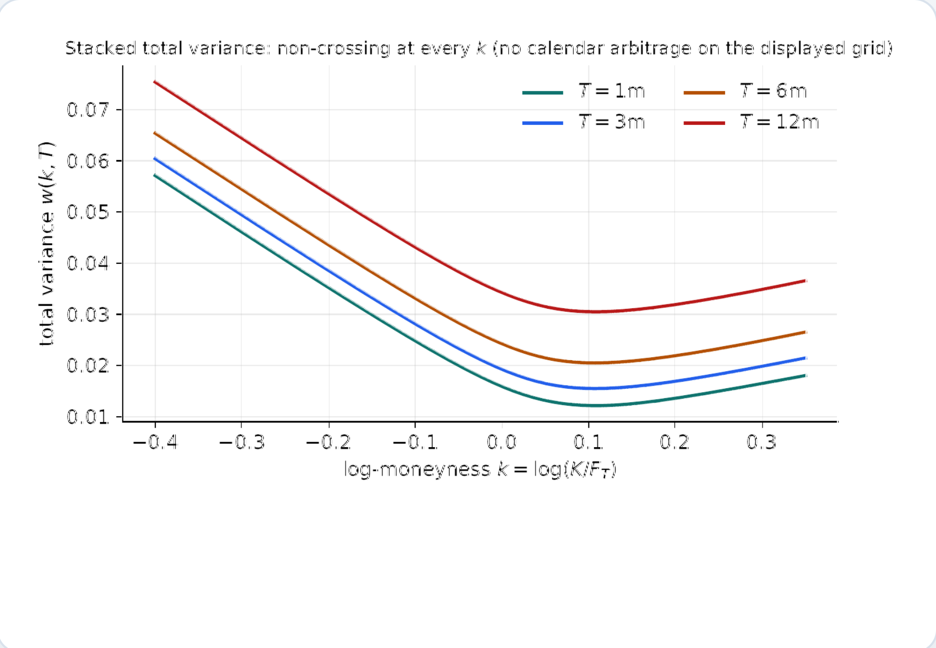
ENGINEERING WHY THE NUMBERS CAN BE TRUSTED

Every claim traces to an equation, a module, and a test — and due diligence is a download, not a promise.

Provenance of everything shown: market-facing screenshots are a **live Massive session**; the graph and filter screenshots are **staged synthetic demos and their captions say so**; backtest numbers are **historical NBBO replay** (Massive/Polygon flat files). Nothing in this deck is a mock-up.

- **1,506 backend + 129 UI tests, green on 2026-07-27** — golden tests pin the code to the notes' equations, byte-identity locks guard every OFF toggle, finite-difference guards every analytic Jacobian.
- **A 22-case certification pack, 22/22 green (report regenerated 2026-07-27)** — every production incident frozen as a named case (market regimes × data failures × model stress) that runs the *same* pytest locks the suite runs, and emits a client-facing HTML/JSON report. Validation and production share definitions instead of drifting apart.
- **16 technical notes + 19 lecture editions** with a fixed contract: boxed central equation, invariant box, traceability table (claim → equation → module → test), executed reference implementation — every figure regenerated by production code, so a drifting number fails the build, not the reader.
- **Two bank-committee-style reviews** (LQD, SVI) sit in the repo verbatim with their triage; the sharpest demand — "*an uncertified slice cannot become a mark*" — shipped as the publish gate. Production case files record what broke, why, and the design rule it produced.
- **A governance kernel:** append-only event log (every quote edit, override and publish, with actor), hash-chained publish manifests (supersede / recall — never delete), and a one-command replay-to-tolerance report, test-enforced at 10^{-9} IV.





Regenerated on every build. Left — all implied densities non-negative: no butterfly arbitrage, by construction. Right — stacked *total variance*, non-crossing at every strike: the actual calendar-arbitrage check, in the same coordinates as the app's Stacked IV tab (measured here on independently fitted slices; the enforcement coupling is Note 10) — produced end-to-end by the production calibrator on the notes' reference surface.

TRADING RELEVANCE

Model governance comes built in: **an auditor can walk from any published surface back to its inputs, its equation, its test — and regenerate the certification report themselves.**

OUTLOOK WHAT REMAINS IS PRODUCTIZATION

The modeling stack is built and validated — what remains is productization, not invention.

- **Recently landed** — the 22-case certification pack (client-facing), the belly-certificate publish gate, the structural SVI chart + analytic Jacobians, the symmetric surface solver, the precision-message propagation default (evidence-gated), the 0DTE research stack (intraday capture, session clocks, degraded mode, 50 ms rail), the joint borrow solve v1, functional bands + closed-form observation planning.
- **Next** — hosted single-tenant packaging (the settled deployment model), daily-horizon graph-band calibration (the recorded ζ~2 fix), the index→single-names intraday campaign, an eSSVI comparator column, a hedge-P&L campaign.
- **Then** — portfolio/P&L views, multi-user data services, enterprise entitlements.

- **Stated limits today** — single-process serving; largest exercised end-to-end: the 25-asset capture universe offline, ~30-node universes live; beyond ~2–3k nodes the sparse graph solver is wanted (deferred deliberately — the message operator is sparse-ready by construction); fully-dark cross-asset skill is demonstrated at assertive reach in stressed regimes, measured ≈0 in calm idiosyncratic tape.
- **Data** — Yahoo (free), Massive/Polygon (delayed + flat-file history), Bloomberg (Terminal) behind one provider contract with health lights and as-of replay; bring-your-own-entitlement by design.
- **Outputs today** — JSON/CSV surface export with a hash-chained manifest, an HTML fit report, the certification and replay reports, and the full REST API; **not yet**: portfolio Greeks/P&L, multi-user entitlements. Deployment: dev server or single-file desktop build.

POSITIONING

A professional-grade fitter plus a graph propagation layer for sparse volatility markets — the fitter has to earn a desk's trust on its own; the graph layer is the part we have not seen offered elsewhere.

VOL-FITTER

One shared handle system — models, priors, filter and graph speaking the same coordinates — so every smile can be marked, certified, and defended.

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